

Selection or More Capital?

Experimental Evidence on High-Growth Entrepreneurs in Kenya

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Abstract

Business plan competitions aim to identify and spur high-growth entrepreneurs. Two experiments were embedded into in a Kenyan competition to test how intensity of selection and of capital determine entrepreneurial outcomes. Applicants received a US\$9,000 grant after a streamlined process or went through multiple selection stages and were randomly assigned US\$9,000 or US\$36,000. All grants

initially generated jobs, but the impacts only persisted over three years under multi-stage selection. The larger grants did not yield greater long-term impacts than the multi-stage US\$9,000 grant, suggesting diminishing returns to capital and limited lumpy investment opportunities. Selection, rather than grant size, determines long-run firm growth..

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1. Introduction

Of the estimated 244 million firms in Africa, 232 million (95%) are own-account businesses consisting of only the owner, and a further 7.7 million (3.1%) have fewer than five employees (Cruz et al, 2025). Efforts to provide finance and training to these microenterprise owners have yielded increases in profits, but zero increase in employment on average (Grimm and Paffhausen, 2015; McKenzie et al, 2025). A key policy objective is to increase the number of high-growth firms that hire workers and expand beyond this micro size. Business plan competitions have become one popular policy tool for this purpose. They aim to identify from the vast pool of current and potential entrepreneurs a subset with the entrepreneurial ability to grow a larger firm, and then to overcome capital constraints on this growth through awarding grants. This raises the questions of how best to select the firms to support, and how much support to provide them. Competitions can use streamlined or multi-phased approaches to selection, and they vary dramatically in the amount of financial support they offer to winners.

A streamlined selection process can save costs and time for both the entrepreneur and government, potentially allowing funding to be accessed more rapidly. It will also be efficient if most of the selection of talent occurs through self-selection into entry, and through eliminating an easy-to-determine bottom tail of entrants. A multi-stage process may increase complexity and delay access to funding with little improvement in outcomes as it can also be very difficult to predict which entrepreneurs will succeed most (Nanda, 2016; McKenzie and Sansone, 2019), let alone which will have the largest treatment effects. However, a multi-stage process may be better if additional steps screen further on ability and commitment and provide a clearer signal of who will benefit most from support.

It is also unclear how much capital is needed to overcome liquidity constraints. A standard concave production function would suggest offering smaller grants to more firms, due to diminishing returns to capital. In contrast, if setting up or growing a firm beyond the microenterprise size requires large indivisible investments, it would be more efficient to

provide larger grants to a smaller number of firms given a fixed budget. In practice there is extremely wide variation in the average size of the grants given in business plan competitions, even just considering competitions in Sub-Saharan Africa. Award sizes range from \$580 in Ethiopia for an ILO Business Plan competition¹ to \$1,000 in the Aspire competition run in Ethiopia, Tanzania and Zambia (Fafchamps and Quinn, 2017); \$5,000 in the pan-African competition run by the Tony Elumelu Foundation²; \$5,000-\$15,000 for a youth entrepreneurship competition in Côte d'Ivoire (Weber, 2014); \$25,000 for a youth employment competition in Mozambique³; \$35,000 for a government competition in Benin⁴; \$50,000 for the YouWin! competition in Nigeria (McKenzie, 2017); and up to \$141,000 in Somalia (Weber, 2014).

We embed two randomized experiments in the context of the *MbeleNaBiz* Business Plan Competition in Kenya in order to test how selection intensity and capital constraints determine the effectiveness of the competition in generating firm growth and employment. The competition was a nationwide competition that attracted more than 12,000 applications. A selection experiment randomizes whether firms go through a streamlined screening, in which only applicants classified with very low potential (26% of eligible applications) are screened out based on a first-stage application form and then submitting a business plan, before 500 firms were randomly selected for \$9,000 grants or for a control group; or a multi-stage screening, in which firms submitted detailed business plans, which were scored and ranked by judges, with the top 750 plans (top 6% of applicants) being eligible to be randomized to a grant or control group. The grant size experiment randomizes these 750 firms to receive either \$9,000, \$36,000, or to a control group.⁵ We then use three

¹ Source: <https://www.ilo.org/resource/article/ilo%E2%80%99s-business-plan-competition-helps-mukhtar-double-his-poultry-business> (grant of ETB 90,000 converted to USD at 2024 exchange rate of ETB 155 /\$).

² <https://www.tonyelumelufoundation.org/tef-entrepreneurship-programme>.

³ <https://agoraemprega.seje.gov.mz/>.

⁴ <https://www.technoserve.org/fight-poverty/projects/strengthening-competitive-businesses-in-benin/>.

⁵ The grant sizes were 900,000 Kenyan shillings (KSh) or KSh 3.6 million. At the time of the call for applications, the exchange rate was approximately KSh 100 to \$ 1, giving the \$9K and \$36K equivalents. By the time the grants were disbursed, the exchange rate was around KSh 107 to \$1, making the grants equivalent to \$8.4K and \$33.6K.

rounds of annual follow-up surveys to measure impacts of this \$13.5 million in grant funding on firm size, employment, and profitability.

We find that the selection process has a large impact on the effects of the grants, whereas the multi-stage \$36,000 (36K) grants do not have significantly higher lasting impacts on employment and firm growth than the multi-stage \$9,000 (9K) grant. In the first year, the 36K grants increased employment by more than 100%, adding an average of 5.0 workers per firm and taking an additional 21.8 percentage points of firms beyond a size threshold of 10 workers from a control base of 11.8%. However, after 3 years, this impact has fallen to 2.6 workers per firm (132% impact), compared to 2.0 workers per firm for the 9K multi-stage grant (102% impact), with both grants having impacts of 9-10 percentage points on getting firms above the 10-workers size threshold. The 36K grant increases monthly sales by \$982, a 69% increase on the control mean, and monthly profits by \$211, a 60% increase over the control mean. The multi-stage 9K grant increases monthly sales by \$839 (59%), and monthly profits by \$183 (52%). We cannot reject equality of these impacts with those of the 36K grant, and we can reject that the 36K grant has four times the impact as the 9K.

In contrast with the multi-stage process, the employment impacts of the streamlined 9K grant dissipate over time, and are only 0.6 workers per firm after 3 years, statistically different from those of the multi-stage grant. The streamlined 9K grant only has a pooled impact of \$79 in monthly sales (7% of the control mean, and not statistically significant), and no impact on monthly profits. We can reject equality of impacts of the two 9K grants.

We find that the large and small grants appear to have been spent proportionally on similar items, and largely for non-lumpy investments in capital and labor rather than large indivisible capital investments. Firms improve their business practices and increase innovation and training, but not much more so with the 36K grant than the multi-stage 9K grant. Therefore, there do not appear to be profitable indivisible investment opportunities that these entrepreneurs can pursue with 36K but not 9K, and the 9K multi-stage grants are much more cost-effective at generating new jobs and boosting firm performance than the 36K grants as a result. The limiting factor on firm size in our sample appears to be entrepreneurial ability,

as in the model of Lucas (1978), and not capital beyond the 9K grant size. The streamlined selection resulted in more women and new firms being selected by design, reweighting for these characteristics does not change the lower performance of these grants. The multi-stage selection does appear to have been better at selecting who can benefit most from the grants.

This paper contributes to several literatures. The first is a broad literature on why there are so many small firms in developing countries and what policy actions may change this. One explanation is that this size distribution is seen as the result of misallocation (Hsieh and Klenow, 2009) due to factors such as size-dependent policy distortions (Restuccia and Rogerson, 2008), informality (Ulyssea, 2018), and financial frictions (Buera et al, 2011). An alternative explanation is that the small size of many firms is optimal given the inherent structural features of low-income economies, such as limited supply of the managerial talent needed to manage large firms (Engbom et al, 2025), small market sizes (Tybout, 2000), and poorer consumers demanding lower quality goods that are optimally produced at smaller scale (Jain and Kothari, 2025). Our evidence that there is a pool of talented entrepreneurs out there who can grow in size when financial constraints are alleviated suggests that the causes of small firms are not entirely due to these structural factors.

Second, our paper contributes to the debate about the extent to which production non-convexities hold back the size of firms and can generate poverty traps. McKenzie and Woodruff (2006) and de Mel et al. (2009) argue that the high returns from small capital investments of \$100-\$200 suggest a lack of non-convexities at low levels of investment. However, Kaboski et al. (2025) find evidence of indivisibilities when comparing \$100 and \$500 grants in Uganda, and Balboni et al. (2022) argue for the presence of a poverty trap investment threshold at around \$500 in rural Bangladesh, although Karlan et al. (2026) show that this conclusion is not robust to functional form choices and may reflect geographic confounds. Our evidence suggests no such indivisibilities within the range of the much larger grant sizes we study.

Third, our paper contributes to the literature on gender and entrepreneurship. Small grants of \$100 to \$200 to women entrepreneurs have had smaller impacts than for men (de Mel et al, 2009 ; Fafchamps et al. 2014; Hussam et al, 2022). This may reflect the efficient scale for many of these firms being very low, given other constraints such as a lack of demand (Hardy and Kagy, 2020) or household constraints being more binding (Jayachandran, 2021). However, there is large heterogeneity in the returns to capital (Hussam et al, 2022; Bryan et al, 2024), suggesting some women may still have high returns. There was no gender difference in impacts of large grants in Nigeria for women making it through the business plan competition (McKenzie, 2017). In our experiments, we cannot reject that the women who won had equal treatment effects from the 36K and 9K multi-stage grants as men. Since women were less likely to be running firms, and had smaller firms in absence of treatment, these similar absolute gains represent larger gains in relative terms for women. The competition thus was able to identify and support the growth of talented women entrepreneurs as well as men.

Finally, we contribute to the literature on business plan competitions as a tool for policy makers to directly identify and spur high-talent entrepreneurs. Key questions revolve around the extent to which high-growth entrepreneurs are capital constrained, and whether their marginal returns to capital vary at different amounts of capital infusion. McKenzie (2017) finds a \$50,000 grant in Nigeria led to firms being 20 percentage points more likely to reach a threshold of 10 or more workers over three years. But grants of this size are expensive to provide, raising the question of whether it would be more effective to support more firms with smaller grants for each. Our results provide support for the hypothesis that funding more firms with less money per firm can deliver higher returns in terms of job creation, but they also show the importance of screening for high returns.

2. The Business Plan Competition

2.1 MbeleNaBiz

MbeleNaBiz (Swahili for “Moving Ahead with Business”) was a business plan competition in the Kenya Youth Employment and Opportunities Project (KYEOP), a US\$150 million World

Bank-financed project, implemented between 2016 and 2023. *MbeleNaBiz* was jointly implemented by the Ministry of ICT, Innovation and Youth Affairs (MIIYA) and the Kenyan Micro and Small Enterprises Authority (MSEA), with technical support from KPMG. The objective was to drive job creation among youth, the age group most likely to be without jobs in Kenya (World Bank, 2016). Women were identified at the design phase as potentially facing additional constraints in starting and growing businesses due to family and household responsibilities (World Bank, 2016), and as being less likely than men to enter competitive settings in lab experiments (Paryavi et al, 2025). This led to a policy interest in examining gender differences in both participation and performance in the competition.

The program was launched in 2019, accepting application submissions between June 24, 2019, and September 2, 2019 (see timeline in Appendix A). The launch was accompanied by a large-scale awareness campaign including radio, internet, newspapers, in-person awareness events in universities, microfinance institutions, incubation centres, and outreach through business associations, women's groups, and tech-development networks. Female participants were specifically targeted in this awareness campaign with the objective of achieving higher participation than in other settings (e.g. in the Nigeria YouWin! Competition, women were 18.5 percent of new business applicants and 14.9 percent of existing business applicants (McKenzie, 2017)).

To be eligible, the lead applicant had to be between 18 to 35 years old, of Kenyan nationality, have completed at least secondary education, and be seeking to operate a business in Kenya. Applicants could apply with either a new business idea or an existing business that they wanted to expand. The program received 12,171 applications, which were screened for eligibility and completeness to 11,466 eligible applications. Twenty-five percent of the applicants were women, and 52 percent were for new business proposals.

Proposals were anonymized and KPMG completed a first-round scoring. Two consultants were assigned to each proposal and scored the quality of the business idea, the expected number of jobs with an emphasis on jobs for young women, what sets them apart from competitors, business risks, the planned investment of the award grant should they win, and

their relevant experience.⁶ The proposals were ranked in multiple buckets of potential with the bottom two buckets (amounting to 26% of eligible applications) screened out during this process, and the top 8,475 applicants randomly allocated to either pass through a streamlined selection process, or go through a multi-stage process, as set out in Figure 1.

2.2 Streamlined Selection Process

A random sample of 1,003 applicants were randomly allocated to the streamlined selection process, stratified by gender and whether the applicant had an existing or new business. This included 500 women and 503 men, oversampling women to have increased statistical power to detect gender differences in effects. We conducted a baseline survey, and those in this group were then asked to submit a business plan online (following the same process and timeline as the multi-stage process described below), with 829 applicants submitting plans. From this set of 829, 250 applicants were randomly assigned to receive a grant of \$9,000, 250 were randomly assigned as a control group, and the remaining 329 had their plans scored in the multi-stage process. Since it was unclear what the composition of awardees would be in the multi-stage process, selection for streamlined awards was stratified by gender and new/existing business status to have equal numbers from each subgroup.

The screening for the recipients of 9k in the streamlined process was then only the initial scoring based on the application form, and the self-selection of completing a business plan. There was no scoring or judging of the business plans to determine these streamlined awards.

2.3 Multi-Stage Selection Process

In the multi-stage process, the scores from the initial applications were used to further reduce the set of applications to the top 4,000 applicants. These applicants were then invited to proceed to a second stage and submit business plans online, to join the 329

⁶ In case of a discrepancy of more than 25% of the score, a third scorer would review the application, and the final score is then computed as the average of the nearest two scores among the three scores.

submissions from the streamlined process.⁷ These business plans were intended to capture the potential of the businesses and their plans for growth, rather than focusing on how they would use the grant money. A total of 3,182 applicants submitted business plans by the deadline of August 31, 2020.⁸ These plans were evaluated by independent judges assembled by KPMG, comprising entrepreneurs, business coaches, and industry leaders. There was an effort to ensure diversity in the judges, and they were a combination of 38 women and 24 men, of whom 11 were international. These judges were each paid an honorarium of around \$500 for contributing their time.

The judges evaluated the business plans on a scale of 0 to 100 based on scoring eleven different elements: the elevator pitch of the overall business plan (6 points); the value proposition (15 points); understanding of customers (7 points); and of competitors (7 points); marketing strategy (15 points); business processes (7 points); human resource plans including potential to create jobs and understanding the roles these workers will play (15 points); financial plans and economic viability (15 points); risk and mitigation strategies (7 points); any additional benefits to society (2 points); and a capstone score for overall promise (4 points). Note that the plans did not say how they would use grants of different sizes, but rather what their overall potential and plans for growth for the business would be.

Business plans were anonymized and scored by one judge, with those that scored more than 60 then moved to a second round to be evaluated by a second judge. Business plans were ranked according to the average of the evaluation scores and used to select the top 750 business plans. A baseline survey of these 750 was conducted in February 2021, before random assignment of the grants.

Screening in the multi-stage process therefore consists of greater selectivity from the first-round application form scores than the streamlined process, self-selection into submitting a business plan, and then selection from judges based on the business plan. The 750

⁷ A random sample of 1,500 of these applicants were encouraged to participate in an online business plan training prior to submitting their plan. Appendix B provides details. Being assigned to training increased the likelihood of submitting a business plan, but not the likelihood of scoring in the top 750 and being a winner.

⁸ This deadline was extended from the original submission deadline due to the COVID-19 pandemic.

represent the top 6% of the initial applicants, according to this multi-stage selection process.

2.4 Grant Size Experiment

The 750 top business plans in the multi-stage process were randomly assigned into three groups: (i) 250 to a control group that did not receive an award; (ii) 250 to the *multi-stage 9K treatment group* to receive a grant of Ksh 900,000 (about US \$9,000); and (iii) 250 to the *36K treatment group* to receive a grant Ksh 3.6 million (about \$36,000). Random assignment was stratified by gender, and whether the applicant had a new or existing business.

2.5 Implementation

The 750 grant winners (250 from the streamlined process, 500 from the multi-stage process) were announced by MSEA on February 12, 2021, and invited to participate in an award ceremony held on February 24, 2021, chaired by the then President Mr. Uhuru Kenyatta.

Awards were disbursed to winners in three payment tranches: tranches of \$6,000, \$20,000 and \$10,000 for the 36K grant; and tranches of \$4,500, \$3,500, and US\$1,000 for the 9K grant. The first tranche was conditional on submitting documents such as a police certificate of good conduct, a credit report, tax identification number, and business registration proof. The second tranche required submitting legal, financial and tax compliance documents along with a progress report, and the third tranche required submitting proof of spending and a progress report. Firms also received monitoring visits before the disbursement of the second and third tranches. Note that none of these tranches were conditioned on business outcomes such as hiring a certain number of workers or showing sales or profit growth. By end August 2022, which was the end of the first follow-up data collection, 741 (99 percent of the total of 750 awardees) had received tranche 1, 720 had received tranche 2 (96 percent), and 662 had received tranche 3 (88 percent).

2.6 Verification of Randomization and Comparison of Firms Selected in Streamlined versus Multi-Stage Process

Table 1 presents summary statistics for applicants in our grant size experiment, and for those randomized to treatment or control through the streamlined process, as well as a

comparison of the 9K grant winners who came through the multi-stage versus streamlined processes. We see that the groups are balanced on baseline characteristics within the grant size and streamlined process separate experiments, whereas there are differences between the winners in the multi-stage and streamlined processes, which is expected given the competition selection. Appendix Table A1 compares the winners in both processes to the other applicants, using data from the application forms.

Applicants are on average 27.5 years old in the multi-stage experiment sample, 27 in the streamlined experiment sample, compared to 26.4 for the non-successful applicants. Twenty-six percent of the multi-stage sample are women and 36% are new business applicants, compared to 50% in each category in the streamlined process due to the stratified selection there. Twenty-three percent of the unsuccessful applicants are women, and 53% were new business applicants. The sample is more educated in the multi-stage competition (80% have at least university education, and 9% postgraduate education) compared to the streamlined process (71% have at least university, and 3% postgraduate education), with both groups more educated than unsuccessful applicants.

Conditional on operating a business, at baseline multi-stage firms have a mean (median) of 6.7 (4) workers, and streamlined firms have 4.8 (3) workers. Monthly profits have a mean (median) of \$540 (\$149) per month for the multi-stage group, and \$418 (\$76) per month for the streamlined group. The 9K grants therefore represent 17-21 months of profits for the average firm, and 5-years or more of profits for the median firm, with the 36K grants then four times this. Only one-quarter of winners with firms had previously borrowed from a financial institution, suggesting limited use of credit for financing growth.

The firms in both experiments span a wide set of industries, with the most common sectors being agriculture (e.g. poultry and livestock production, drip irrigation systems to enable multiple growing seasons); wholesale and retail trade (e.g. sale of clothing and fashion accessories, cereals and food items, and cosmetics and beauty products); information and communication technologies; and manufacturing (e.g. making alternative energy products

from agricultural waste, grain milling, food processing). The sectoral mix is similar for the multi-stage and streamlined processes, with these firms more likely to be in agriculture and manufacturing, and less likely to be in trade or ICT than the unsuccessful applicants in Table A1.

3. Measuring Impacts

3.1 Data Collection

Baseline surveys were conducted before random assignment of the grants, in February 2020 for the streamlined process group, and in January 2021 for the multi-stage group. This difference in timing reflects a pause during the start of the COVID-19 pandemic, as well as the time taken to go through the business plan selection to score and rank the multi-stage sample applicants. Note that although the timing of the baseline surveys differed, the timing of awarding and receiving grants did not differ between these two selection approaches. We measure impacts on firm outcomes through three rounds of follow-up surveys. These surveys were generally conducted by face-to-face interviews, with shorter phone surveys administered to respondents who could not be interviewed in person. We collect data on primary outcomes related to employment in the firm, and firm performance, as well as data on mechanisms such as investments in capital, the business practices and innovation activities undertaken by the firm, and the non-business work activities of the applicant. Appendix D discusses how our different outcomes are constructed.

The first follow-up survey (FU1) was conducted between May and August 2022. It was purposely implemented right at the end of the main disbursements to assess how the beneficiaries used the grants, capturing impacts in the first year. The second follow-up (FU2) took place between February and May 2023, measuring impacts approximately two years after the first disbursements. The third and final follow-up survey (FU3) was conducted between July 2024 and May 2025, measuring impacts 3-4 years after the first disbursement.⁹

⁹ This longer window for conducting the final follow-up reflects additional time and effort devoted to trying to reduce attrition.

3.2 Estimation of Treatment Impacts

We stack the two experiments and estimate the following ANCOVA specification for outcome Y_{ist} for entrepreneur i in randomization strata s , separately for each follow-up survey round $t \in \{1,2,3\}$:

$$Y_{ist} = \alpha + \beta_1 TREAT36K_i + \beta_2 TREAT9KMULTI_i + \beta_3 TREAT9KSTREAM_i + \lambda_1 Y_{i0} + \lambda_2 (MULTI_i \times Y_{i0}) + \pi_1 M_{i0} + \pi_2 (MULTI_i \times M_{i0}) + \rho MULTI + \delta_s + \varepsilon_{ist}$$

where $TREAT36K$ denotes assignment to the 36K grant, $TREAT9KMULTI$ denotes assignment to the 9K grant under the multi-stage process, and $TREAT9KSTREAM$ denotes assignment to the 9K grant under the streamlined process. β_1 and β_2 then identify the impacts of the multi-stage grants relative to the multi-stage control group, while β_3 the impact of the streamlined grant compared to the streamlined control group. We control for the baseline value of the outcome where available (Y_{i0}), and dummy this out if missing (M_{i0}), allowing the relationship to differ depending on whether the respondent is part of the multi-staged process (dummy variable $MULTI$). The δ_s denote randomization strata fixed effects, and ε_{ist} is the error term. We use robust (Eicker-White) standard errors.

In addition to being interested in the direct effects of the three treatments ($\beta_1, \beta_2, \beta_3$), we also test the following three hypotheses: (1) the 36K and 9K multi-stage process grants have the same effect ($\beta_1 = \beta_2$); (2) the 36K grant has four times the impact of the 9K multi-stage grant ($\beta_1 = 4\beta_2$); and (3) the two 9K grants have the same treatment impacts ($\beta_2 = \beta_3$).

We estimate the impacts on primary business outcomes round by round, since we are interested in the trajectory of impacts. We test for equality of impacts across time and consider a pooled specification to increase power. The pooled specification stacks all three follow-ups, including survey round fixed effects, and clusters the standard errors at the firm level.

Our AEA registration specified the primary outcomes as employment and profitability, in keeping with the policy objectives of the program to select and support well-performing firms that could generate jobs for others. Since our sample includes individuals who did not have a firm at the time of application, and firms shut down over time, this raises the question of how best to code outcomes for applicants without a firm. For our main specifications, we follow standard practice in the literature of examining unconditional impacts, which code an applicant without a firm as having 0 employees and 0 profits. Our treatment impacts then capture the combination of the extensive margin impact (whether they run a firm at all) and intensive margin impacts (how big and profitable these firms are). We also measure separately the extensive margin impact on operating a firm, and examine descriptively the intensive margin impacts conditional on operating a firm. These conditional impacts combine both a treatment effect and any selection effect of how the grants change who operates a firm, and we discuss the likely biases in interpreting those results.

3.3. Dealing with Survey Attrition

The survey response rates declined over time, from 83% in the first follow-up, to 68% in the second, and 64% in the third, with attrition rates higher for the control groups than treatment groups, particularly in the multi-stage experiment (Table C1). Appendix C details our efforts to reduce attrition. Attrition mostly arises from refusals, rather than being able to find applicants. We can ascertain whether the business is operating or not for many of these refusals through direct and indirect responses. In the third follow-up round, 70% of those in the multi-stage process control group who completed the survey were operating a business, compared to only 51% of those who did not answer but whose status was known. This suggests that our treatment versus control comparisons may be lower bounds, since we are missing more closed firms from control than treatment. Our endline sample remains balanced on most baseline observables (Table C2). After showing our main results, we examine robustness to controlling for baseline variables,¹⁰ using different imputation assumptions for the attritors, and to bounding approaches. Our results are robust overall to

¹⁰ Following Cilliers et al. (2025), we include the full set of baseline controls rather than using PDS Lasso given the relatively small set of controls and our sample size.

these different assumptions. It would require that the control attritors were all very successful entrepreneurs – an assumption at odds with the data – to overturn our treatment estimates.

4. Impacts on Firm Operation, Employment, and Firm Performance

4.1. What does theory predict we should see?

We provide a simple organizing framework to illustrate how our two experiments in the business plan competition might be expected to affect firm outcomes. Given a production function $f(\cdot)$, firm productivity A , and entrepreneurial skill E , the firm owner will choose the amount of capital K and labor L to maximize profits π , subject to a borrowing constraint B :

$$\max_{K,L} \pi = pf(A, K, L, E) - wL - rK \quad s.t. \quad K \leq B$$

Where we assume the firm is a price-taker, facing output price p , wage rate w , and interest rate r . The first-order conditions then set the value of the marginal product of labor equal to the wage rate, and the value of the marginal product of capital equal to the shadow cost of capital (where λ is the Lagrange multiplier on the borrowing constraint):

$$pf_L(A, K^*, L^*, E) = w$$

$$pf_K(A, K^*, L^*, E) = r + \lambda$$

We see that if firm owners are not liquidity constrained, then an unconditional grant will not change these first order conditions, and would therefore be predicted to have no impact on either capital or labor. This is the standard additionality concern - grants may just fund the activities that firms were already planning on doing (crowding out other investment), and the treatment effect would then be zero. McKenzie (2017) notes that when business plan competitions pay money in tranches, this can be thought of as a conditional grant, which temporarily increases the returns to investing in capital and labor. This would cause entrepreneurs to overinvest in their firm, and have a temporary treatment effect, but this would then reverse once all tranches are paid. In order for the grants to have a lasting impact on firm size in this framework, we therefore need liquidity constraints to bind, so that the firm is operating below its efficient size.

Impact of Streamlined versus Multi-Stage Selection: As in the model of Lucas (1978), optimal firm size will depend on entrepreneurial ability E . If we assume that E is a complement to K and L , then all else equal, more able entrepreneurs will have a higher steady state firm size, and so potentially be more likely to be credit-constrained and to benefit from treatment.¹¹ To the extent that the multi-stage business plan screening results in higher quality entrepreneurs than the streamlined screening, we would therefore predict larger treatment effects from the multi-stage. Alternatively, if the more intensive screening does not result in higher ability or more constrained entrepreneurs, then we should not expect to see different treatment effects from the streamlined selection process as from the multi-stage process.

Impact of Grants of Different Sizes: How will the treatment impact vary for a grant of $G_1 = \$9,000$ compared to a larger grant of $G_2 = \$36,000$? The answer will depend on the shape of $f(\cdot)$. The standard assumption in the literature is that $f(\cdot)$ is concave, so that there are diminishing returns to capital, which leads to the first-order conditions above solving for a global maximum for optimal firm size. Then the marginal return to additional grant funding will fall as more funding is added for two reasons. First, more funding makes it less likely that the borrowing constraint still holds, and once this occurs, $\lambda = 0$, and we are back in the unconstrained case above, where further grant funding beyond what is needed to alleviate the borrowing constraint is not invested in the firm, just crowds out other investment, or is temporarily invested due to the tranche structure, but then decapitalized over time. Second, even if the borrowing constraint still binds, $f_{KK} < 0$ means that the per-dollar treatment effect from G_2 will be less than that for G_1 .

An alternative prediction would occur if the production function $f(\cdot)$ is non-convex, such as when there are large indivisible investments needed. Then there can be multiple steady states, with entrepreneurs with low levels of wealth stuck in a poverty trap (Banerjee and Newman, 1993; Buera, 2009). Then it is possible that the smaller grant size G_1 may be

¹¹ This assumes that E does not affect access to credit. If more able entrepreneurs are also better at attracting credit, the borrowing constraint may bind less strongly for them, and we should expect smaller treatment effects. Higher ability entrepreneurs could have also saved more in anticipation of entering entrepreneurship, as in Buera (2009).

insufficient to enable purchase of the large asset, and so entrepreneurs receiving the small grant may have low marginal returns. In contrast, if the larger grant G_2 can fund purchase of the lumpy asset, then this will have much higher returns, and the per-dollar treatment effect from the large grant could be greater than that of the small grant.

This discussion assumes that the grant only affects the profit maximizing decision and optimal firm size through affecting the borrowing constraint on capital. It is possible that the borrowing constraint may also directly affect the ability to hire labor if it takes time for firms to train workers to be productive, and if norms or labor laws require paying workers a wage above their marginal product during this period. The grant could also potentially be used to increase E , if owners invest it in business training, or A if they hire consultants or invest in innovation. Finally, the model above assumes that firms are price-takers, and can sell all the output they can produce at price p . However, in practice a lack of demand is often a key constraint on firm growth (e.g. Hardy and Kagy, 2020), and investments in marketing and customer acquisition may allow the firm to grow by being able to sell more of its output or increase the price of its output. This suggests examining whether the grant is spent largely on capital and labor inputs, or whether it is used for these other activities.

Note further that this framework abstracts from risk, but the inherent riskiness of entrepreneurship and high-growth activities is one key reason why loans are unlikely to be available to fund these activities. In theory, micro-equity would be another potential policy instrument, but the difficulty of monitoring profits and enforcing repayment for firms of this size can make implementation challenging (de Mel et al, 2019). Risk aversion could be another reason the larger grant is used differently from the smaller grant, since the additional funding can help the entrepreneur self-insure against an investment of a given size not paying off. We will examine how the grants are spent to see if the larger grants disproportionately go to riskier activities.

4.2 Empirical Impacts

4.2.1. *Impacts on Jobs*

Table 2 shows that all three grants increased the likelihood of an applicant operating a business, led to more workers being employed in these businesses, and resulted in more firms getting to a size of 10 or more workers. Panel A examines the extensive margin of running a business at all. In the first follow-up round, 83% of control applicants in the multi-stage selection experiment, and 74% of controls in the streamlined selection experiment are running a firm. The treatment effects are 14-16 percentage points in the multi-stage experiment, and 22 percentage points in the streamlined, so that 96% or more of treated applicants were running a firm. We cannot reject that the treatment effects remain constant over time in the multi-stage experiment, with those offered the 36K grant being 14.3 percentage points more likely to run a firm 3-4 years later, and those given the multi-stage 9K grant being 11.3 percentage points more likely. In contrast, the impact of the streamlined 9K grant falls over time, from 22.3 percentage points after one year to only 7.9 percentage points after 3-4 years. Nevertheless, we cannot reject that the two types of 9K grant have similar long-term effects, nor can we reject that the 36K grant has the same impact as the 9K.

Panel B of Table 2 shows the impact on job creation in these firms, and Panel C on whether firm size reaches 10 or more workers. Control group firms in the multi-stage experiment are employing an average of 3.9 workers in the first follow-up, which falls to 2.0 workers in the third follow-up. These are unconditional means, which code firms that close as having zero workers, and so the drop in workers in part reflects firms closing over time. Of the control firms, 11.8 percent had 10 or more workers at follow-up 1, falling to 5.4 percent in follow-up 3. The large grant of 36K has a larger short-term impact on employment than either of the 9K grants, more than doubling firm size by adding an average of 5.0 workers (131% over the control mean) per firm in follow-up 1, taking an additional 21.8 percentage points of firms beyond the 10-worker threshold. This compares to a 2.6 worker treatment impact of both 9K grants in the first follow-up (67% impact for the multi-stage 9k and 97% for the streamlined 9k). While this impact is statistically larger, we can reject that the larger grant leads to four

times the increase in employment as the smaller multi-stage grant. Moreover, while the impact of the multi-stage 9K grant is not statistically different across rounds, still averaging 2.0 workers after 3-4 years, the impact of the 36K grant falls over time. In the third follow-up it is 2.6 workers, which is not statistically different from that of the 9K multi-stage grant, and likewise the impact on having a size of 10 or more workers of 9.7 percentage points is not statistically different from the 8.6 percentage point effect of the 9K multi-stage grant. In contrast, while the streamlined 9K grant has similar short-run impacts on employment and firm size to the multi-stage grant, these impacts dissipate over time, and are small (0.6 workers), not statistically significant from zero, and statistically different from the multi-stage 9K grant in the last follow-up.

We explore further the changes in labor inputs in Table A2. The 5.0 worker short-term increase in employment from the 36K grant is almost evenly split between a 2.3 worker increase in wage workers, and 2.4 worker increase in paid casual workers, with little usage and no change in unpaid workers. The increase in wage workers is constant across rounds, whereas the impact on casual workers falls. We cannot reject that the 36K grant has the same long-term impacts on both wage workers and casual workers as the multi-stage 9K grant, while the streamlined 9K grant has only a lasting long-term impact on casual workers (of 0.6 workers) and not on wage and salaried workers (0.1 workers). In addition to more paid labor, firm owners devote more of their own labor hours to their firms with all three grants. Applicants receiving the 36K grant work 14 hours more a week in their firms in the first follow-up (relative to a control mean of 29 hours), which falls to 6 hours more a week by the 3-4 year follow-up. Applicants selected for both types of 9K grants average 7-9 hours more per week across the three follow-up rounds.

We also examine whether there are differences in the types of workers hired in Table A3. Our first and second follow-up surveys ask about the highest paid worker hired in the last year. The average highest paid worker hired is 29-30 years old, 30%-40% are female,¹² and over

¹² For female entrepreneurs, the share reporting their highest-paid employee as female is 10 percentage points higher.

85% are not related to the owner. The 36K grant leads to more educated workers being hired in the short-term, and who are more likely to be found through advertisements rather than family networks, but who are not paid any higher wages than workers hired in the other treatments. It is thus not the case that the larger grants seem to be going to share rents through hiring less skilled relatives. In the last follow-up round we ask about the most recent hire, and the typical worker, and see few differences across treatment groups nor with the control group. The grants therefore do not appear to have changed the long-term composition of employees.

4.2.2 Impacts on Firm Performance

Table 3 shows that only the 36K grant and multi-stage 9K grant have significant and sustained impacts on firm sales and profitability, whereas the streamlined 9K grant does not. We cannot reject that the impacts are constant over time across the three follow-up rounds for any of the three treatments, and the magnitudes are reasonably similar across rounds, so we focus discussion on the pooled results. The 36K grant increases monthly sales by \$982, a 69% increase on the control mean of \$1,424, and monthly profits by \$211, a 60% increase over the control mean of \$352. The multi-stage 9K grant increases monthly sales by \$839 (59%), and monthly profits by \$183 (52%). We cannot reject equality of these impacts with those of the 36K grant, and can reject that the 36K grant has four times the impact as the 9K. The impact on a standardized index of profits and sales is 0.16 s.d. for 36K and 0.14 s.d. for multi-stage 9K. In contrast, the streamlined 9K grant only has a pooled impact of \$79 in monthly sales (7% of the control mean), and \$-18 in monthly profits (6% of the control mean), none of them statistically significant (compared to the control). We can reject equality of impacts of the two 9K grants.

Earnings are also higher conditional on operating a firm and when looking at total labor income including wages. The impacts shown in Table 3 are unconditional, and so higher sales and profits can come from applicants being more likely to run firms, as well as from earning more in these firms. In Table E1 we show treatment-control differences on the intensive margin. The point estimates show a \$729 increase in monthly sales and \$149

increase in monthly profits for the 36K grant, and \$529 increase in sales and \$124 increase in profits from the 9K grant. These are estimated less precisely and only represent true treatment effects under the assumption that treatment does not induce selection into who runs a firm. But they are suggestive of firms earning more conditional on operating. A related question is whether running a firm crowds out better earning opportunities in the labor market. Table E2 shows this is not the case. The impact on total work income, which includes profits from the main business, from any secondary businesses, and wage income, is \$223 per month from the 36K grant (44% increase on the control mean), and \$260 per month from the multi-stage 9K grant (51% increase), compared to a slightly negative and insignificant point estimate from the streamlined 9K grant.

Labor productivity is not significantly higher as a result of the grants, although it is measured imprecisely and may in part reflect selection into operating a firm. Firms with the 36K and multi-stage 9K grant have more workers and are selling more output. Table E2 shows that log sales per worker is not significantly different between treatment and control in the multi-stage process and is in fact significantly negative for the streamlined 9k grant. The point estimate is -0.07 for the 36K grant, and -0.02 for the multi-stage 9K grant. However, the confidence intervals do include up to 16 to 23 percent improvements in labor productivity. If the extensive margin effect resulted in getting some lower productivity firms to operate (which would not have done so absent the grants), these labor productivity estimates will be downward biased.

4.2.3 Robustness of Main Results to Attrition

Table 4 shows our main results on jobs, firm size, and profitability are robust to multiple ways of dealing with attrition. We focus on impacts in the third follow-up, which is when the attrition rate was highest. Column 1 shows our treatment estimates from Table 2 and Table 3. Column 2 controls for any selective attrition on the 27 baseline variables presented in Table 1. Columns 3 to 5 follow Campos et al. (2025) in assessing robustness to three different imputation approaches for attritors. In keeping with their lower likelihood of operating a firm (see Appendix C), Column 3 assumes that all firms not interviewed earn zero profits. Column

4 assumes firms that attrit have the maximum number of employees or profits that they have ever been observed to have before in previous rounds (including baseline). Column 5 assumes that firms that attrit are like those that are hard to interview, giving them the average outcome for firms that only answer one of the three follow-up rounds. We see that the results are qualitatively similar across all these specifications, with the magnitudes not varying greatly. For example, the impact on having 10 or more workers ranges from 0.07 to 0.10 for the 36K multi-stage grant, 0.06 to 0.09 for the 9K multi-stage grant, and 0.01-0.03 for the streamlined 9K grant. Columns 6 and 7 construct Lee bounds. Column 6 constructs Lee bounds for comparing the treatments to control. Here, the lower bounds in the multi-stage grants are negative for the likelihood of getting over 10 employees or profits, so our treatment estimates would be overturned if it were the case that the additional attritors in the control were all the most successful firms. This is not what our data suggests from both the baseline data and from seeing that firms that attrit are more likely to be closed. Column 7 instead constructs Lee bounds for comparing the two multi-stage grant treatments to one another. Since attrition rates were similar for these two treatments, these bounds are narrow, making the conclusion of similar treatment effects robust.

4.3 Heterogeneity in Treatment Impacts

A key targeting question in running business plan competitions is whether impacts will differ by gender and by whether the applicant already has an existing business. This helps inform whether there is an equity-efficiency trade-off in trying to support new firms and women. Women were only 25% of applicants to the competition and were 26% of winners in the multi-stage process. Table 5 shows that we cannot reject that the women who did win had equal treatment effects for both the 36K and 9K grants as the male winners. We consider the pooled impacts across the three rounds to maximize power, and find similar impacts by gender on employment, firm size, and profits. Women are less likely to be running a firm, and have smaller firms in the absence of treatment, so these treatment effects are larger in relative terms for women. For example, only 1.1 percent of women in the multi-stage control group reach 10 or more workers, compared to 10.0 percent of men. The treatment effect of the 36K grant of 14.7 percentage points for women is a 13-fold increase, compared to the

14.3 percentage point increase for men being a 1.4-fold increase. We likewise cannot reject equality of impacts by gender of the streamlined 9k grants – of which half were allocated to women, and the point estimates tend to be larger for women. This suggests that the lower impacts of the streamlined 9K grants are not due to the higher proportion of women (we show this formally by reweighting the data in section 5.2).

Individuals with existing businesses comprised 48% of applicants, but made up 66% of winners of the multi-stage process. Table 6 shows that there is only limited heterogeneity in treatment effects from the grants by pre-existing business status, which is not robust to multiple testing corrections. The 36K grants have significantly larger (at the 10% level) impacts on firm profits for existing firm owners than new business applicants: \$328 per month in profits compared to \$21 for new business applicants. In contrast, the gap in firm performance impact by pre-existing business status is much smaller for both types of 9K grants and not statistically significant. Nevertheless, we still cannot reject equality of impacts of the 36K and 9K grants even for those with existing businesses. Table A5 and Table A6 also show we do not find any consistent and significant heterogeneity in treatment effects for firms located in a larger city (where we might think demand constraints bind less), nor with whether the entrepreneur has at least college education (which might proxy for ability).

Figure 2 examines how the 3-4 year treatment effect varies over the distribution, showing that the 36K and 9K multi-stage grants have similar impacts to each other on employment and profits at almost all quantiles. The quantile treatment effects increase with quantile. The impacts on employment are zero at the 20th percentile or below, reflecting firms not being open. They are then positive and significant above this percentile, growing from less than 1 additional worker at the 30th percentile to over 5 workers at the 90th percentile. The streamlined 9K grant has smaller impacts everywhere. Quantile treatment effects on profits show similar patterns to that for employment, but with less precision. Treatment effects are again highest at the top quantiles for the multi-stage grants, but not for the streamlined 9K grant.

4.4 Job Creation and Cost Effectiveness

We can use our treatment effects to estimate the total number of jobs created by this business plan competition, and the cost per job or per job-year.¹³ The 750 winners created a total of 2,555 additional jobs in year 1, 1,769 in year 2, and 1,313 in the 3-4 year follow-up. The cost per job created in the third follow-up is \$13,672 for the 36K grant, \$4,438 for the 9K multi-stage grant, and \$15,228 for the 9K streamlined grant. By way of comparison, McKenzie (2017) calculated a cost of \$9,600 per job in year 3 created by the grants averaging \$50,000 in the Nigerian YouWin! business plan competition. Under the extremely conservative assumption that none of these jobs remain after the 3 year follow-up, the cost per job-year is \$3,274 for the 36K grant, \$1,340 for the 9K multi-stage grant, and \$1,861 for the 9K streamlined grant. For comparison, the third follow-up survey collects data on the monthly wages of the most recent hire and typical employee in these firms. Wages have a median of \$100-\$116 and mean of \$119-\$146 per month.

5. Why Do We See These Impacts?

We find that the 36K grants do not yield significantly higher impacts on firm growth than the 9K multi-stage grants, and that the 9K streamlined grants have lower impacts than the 9K multi-stage grants. This section explores different explanations for these results.

Is it that they spend the money differently, diminishing returns to capital, or that selection identifies people who have higher returns to given spending?

5.1. Did the way the grant was spent and invested differ across groups?

Our framework noted that the extent to which we should expect the 36K and 9K grants to differ will depend on the shape of the production function (and whether the bigger grant can finance indivisible investments that the smaller grant cannot), as well as on how binding

¹³ McKenzie (2017) discusses the assumptions needed for these calculations and benchmarks against other programs. The key assumption is that these jobs represent new employment, rather than just crowding out hiring that other firms would have done instead. This may be plausible in an environment with a limited number of SMEs in which many young people struggle to find wage jobs.

liquidity constraints are (and hence whether the grants crowd out other investments the firm would otherwise have). We look at how the grants were spent to examine this.

The large and small grants appear to have been spent proportionally on similar items, and largely used for non-lumpy investments in capital and labor. Figure 3 summarizes the direct reports, based on the most recent follow-up in which this information was collected (first or second), of how grantees say they spent the money. The largest single category is money spent on raw materials and inventories, which accounts for \$10,500 of the 36K grant (29%), and \$2,900 of the two 9K grants (32%). The next two biggest items are spending on building or office space, which appears likely to be mostly rent rather than purchase (see below), accounting for 13%-16% of each grant type; and paying employees (9%-12%). We see only small shares of spending allocated to lumpy assets (3%-6% on vehicles, 2%-5% on land, 1% on livestock). Collectively 14%-19% is spent on items that might affect A and E in our model: formalization, new technologies, marketing, and research and development. Finally, 8%-16% of the grant was still unspent (which could reflect just having recently received the last tranche for some people). These expenditure patterns do not suggest that the 36K grants were invested in different or higher risk activities than did the 9K grantees.

Money is fungible, and so reporting that the grant was spent on these categories need not result in total spending on these items increasing if the grants crowd out other investment. Table 7 uses our surveys to measure treatment impacts on capital stock and use of finance. Our first-round survey asked for the value of *owned or rented* capital stock, while the third round asked only for *owned* capital stock. Column 1 shows very large increases in land and buildings owned and rented in year 1, with the point estimates much larger than the direct reports of how money was spent, and larger than the total amounts of the 9K grant. Even after winsorizing at the 99th percentile, this can reflect the influence of the top tail,¹⁴ as well as potentially crowding in own investments. But column 2 shows much smaller impacts on

¹⁴ The quantile treatment effect at the median is \$4,300-\$5,700 for the two 9K grants, and \$14,000 for the 36K grant.

land and buildings owned in year 3, suggesting that much of the spending was on rent, rather than purchase.

Inventories increase by \$4,800 after 1 year for the 36K grant, and \$4,300 for the 9K multi-stage grant, with these impacts falling to \$1,600 for both groups in year 3. Other capital stock excluding land, buildings, and inventories increases is \$7,300 after year 1 for the 36K grant and \$5,000 after year 3 (compared to \$12,500 and \$3,100 in the control group, respectively), while there is no significant impact for the multi-stage 9k grant. There is no lasting significant impact on capital stock for the streamlined 9K grant.

Approximately 30% of firms take a loan in any given year, and treatment does not significantly change this. There is a positive impact of the 36K grant of \$1,672 on the amount borrowed, and no significant impacts on amount of borrowing from the two 9K grants. So the grants do not appear to be crowding out external financing.

Table 8 shows positive impacts of the grants on business practices, having a website, and on an index of innovation and training, suggesting that A and E have improved. Innovation into new products and markets allows firms to expand without just competing away sales from other firms. We consider both unconditional impacts, which code the outcome as 0 for those not operating firms, as well as differences conditional on operating a firm. Business practices include 26 questions that cover marketing, buying and stock control, record-keeping, and financial planning practices taken from McKenzie and Woodruff (2017). Treated firms in the multi-stage selection process are doing 18-22 percentage points more practices, and 5-8 percentage points more conditional on being in operation. They are 17-18 percentage points more likely to have a business website (relative to a control mean of 19% conditional on operation). Our innovation and training index is an average of standardized z-scores of 20 components (shown in Table A4) that cover innovation, training of workers, participation in training, and use of the internet for business. The grants increase this index by 0.13-0.18 s.d. unconditionally, and 0.06-0.10 s.d. conditional on operating. This comes from firms being more likely to introduce new products and processes, more likely to both attend training themselves as well as provide training in new areas to their workers, and more

likely to use the internet for a variety of business tasks. These improvements in business practices and in innovation are significantly larger from the 36K grant than the multi-stage 9K grant, although the magnitudes of these differences is not large (3-4 p.p. conditional on operating). The streamlined selection 9K grant has smaller point estimates that are statistically different from the multi-stage selection 9K grant for business practices conditional on operating, and for having a business website unconditional on operating.

These improvements in business practices and new product innovation also suggest that the treatment effects do not just come from business stealing from other firms. Prior work has found that when firms are all located together in the same cluster, and only some get credit which they use to just do more of the same activity, these firms grow at the expense of competitors (Cai and Szeidl, 2024). In contrast, the market as a whole can grow if firms innovate through introducing new products (McKenzie and Puerto, 2021). Increasing convenience and product variety can allow gains to come without taking business from incumbents (Bronnenberg and Ellickson, 2015). Our firms are spread throughout the country. We only have GPS data for approximately half the sample, with the median firm 7.3 kilometers from the closest other firm in the study in the same broad (e.g. manufacturing) sector. Treated firms are thus unlikely to be growing through taking sales from control firms. The shortage of firms with 10 or more workers in the country, coupled with selling new products to the local area suggests treated firms are also likely to not just be taking sales from non-study firms.

Taken together, this evidence helps explain why we do not see the 36K grant yielding bigger lasting impacts on job creation and firm profitability than the multi-stage 9K grant. It appears that the 36K group spent a lot of their grant on short-term inputs like rent, raw materials, and worker salaries. This temporarily increased firm size, but did not generate much more in profits than with the 9K multi-stage grant. The 3-4 year impacts on K, A, and E were similar for the two grant sizes, and appear likely to have taken the average firm to a point where the budget constraint was no longer binding, resulting in the same level of L and hence of production and profits. This suggests that these entrepreneurs were not facing lumpy

investment opportunities between 9K and 36K that were only possible with the larger grant,¹⁵ and that they were not able to find enough new innovative activities with the large grant to increase the optimal size of capital stock and labor beyond what was possible with the smaller grant. This suggests that the limiting factor on firm growth may be E, the entrepreneurial capacity of the grantees, and for larger impacts the competition would have had to either attract and select more innovative entrepreneurs, or else provide capacity-building support that greatly increased E.

5.2 Do observable characteristics explain the differences in 9K grant impacts?

Our results find that the streamlined 9K grant was spent in a similar way to the multi-stage selection 9K grant, but had smaller longer-term impacts on firm size and firm performance. Its impacts on labor and capital were also much more similar in the first year to the multi-stage selection grant than in the longer term. This suggests that these participants were in the scenario in which they were not that credit-constrained, so that the grants crowded out other investment they would have made, and they temporarily grew beyond their optimal scale and then shrank back. But this then raises the question of why these individuals had lower returns to the grants.

Table 1 compared the baseline observable characteristics of the two types of 9K grantees. We see the individuals selected through the streamlined process differ in several key aspects from those selected in the multi-stage process. The first is in terms of gender and existing business ownership. The streamlined process stratified on these characteristics, and ended up selecting more women, and more new business applicants than the multi-stage selection. Tables 5 and 6 showed limited heterogeneity in treatment effects by these characteristics, suggesting that they are unlikely to explain the difference in treatment effects across the two types of 9K grants. To examine this formally, in Table 9, panel A, we reweight the streamlined 9K grantee sample so that it has the same distribution of gender and new versus existing businesses as the multi-stage 9K sample, and report impacts on the

¹⁵ Of course we cannot say whether there are profitable indivisible investment opportunities that could only be pursued with much larger grants beyond the 36K amount.

outcomes pooled across waves. Reweighting does not make much difference to the estimated impacts of the streamlined 9K grants, which remain smaller than those of the multi-stage 9K grant.

Table 1 also shows that the streamlined 9K grant awardees differ in other key observable characteristics from the multi-stage 9K awardees. They are less educated on average (68% have college education compared to 82% in the multi-stage group) and earn less income from work at baseline, and conditional on running a business, have lower sales and less capital. These differences are consistent with the streamlined process selecting less experienced and less capable business owners on average, who have a lower E , and hence lower optimal scale. In Table F1 we examine heterogeneity in impacts of the 9K streamlined grant for those scoring above versus below the median on the application scores for this sample. We do see business outcomes are higher for those with above the median scores, but the sign on the interaction with treatment is negative and we cannot reject that treatment impacts are the same. Panel B of Table 9 uses inverse propensity scores to reweight the sample based on the application score, and we see that making the two samples more comparable on first-round score also does not explain the difference in impacts of the 9K grant between streamlined and multi-stage firms.

This suggests the gains are coming from selection on the business plan scoring, rather than just from being more selective in terms of the first-round application scores. It therefore seems that the multi-stage screening and judge scores are playing an important role in identifying entrepreneurs with larger treatment effects. Figure F1 shows that there is only a very weak correlation (0.062) between the first-round application scores and judge scores for the sample who submit business plans, so that the detailed business plans and judge scores are providing independent information. Table F2 shows the interaction of treatment with this score is positive even within the subsample of the top 750 randomized to the multi-stage grants, although only significant for profits in the pooled sample among the 36k. Table F3 examines what characteristics are associated with higher judge scores. Judges score firms higher if they are in manufacturing, have an existing firm, have higher sales, and are

registered. But the R^2 is only 0.09, and it therefore appears that there is a lot that the judges' scores are picking up that is not captured by observable data.

6. Conclusion

One view of the large number of single-person and tiny firms in developing countries is that this reflects the efficient scale for serving poor consumers in fragmented markets, and that larger firm sizes will only come as a consequence of economic development. An alternative view is that this firm size distribution reflects misallocation, and is thus one of the causes of underdevelopment, as talented entrepreneurs cannot access the resources they could use to grow bigger firms.

Our results suggest that the current firm size distribution in Kenya is not completely due to there being no opportunities to grow larger firms. A business plan competition that can identify talented entrepreneurs who have the ability to grow their firms to ten or more workers and financing them with capital grants leads to sustained increases in firm size. However, our results also show that it matters how and who you select to support, and suggest limits to how much capital their businesses can absorb.

A multi-stage selection process that selects the very top applicants does better than a streamlined process that just removes the lower tail of applicants, but giving a grant of \$36,000 does not deliver significantly higher returns than providing \$9,000. This suggests that these smaller grant sizes can allow business plan competitions to support more entrepreneurs or be implemented more cost effectively. More research is needed on policies that can unlock productive opportunities for firms to absorb higher amounts of capital, such as by linking large grants to tailored training or consulting, or connections to larger markets.

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Tables

Table 1: Balance at Baseline

	Control-Multi mean	36K mean	9K-Multi mean	P-value: Control-Multi = 36k = 9k-Multi	9K-Stream mean	Control-Stream mean	P-value: 9k-Stream = Control-Stream	P-value: 9k-Multi = 9k-Stream
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Strata Variables								
Female - Existing Business	0.184	0.172	0.188	0.890	0.256	0.256	1.000	0.068
Female - New Business	0.076	0.088	0.072	0.788	0.240	0.240	1.000	0.000
Male - Existing Business	0.472	0.456	0.464	0.938	0.260	0.260	1.000	0.000
Male - New Business	0.268	0.284	0.276	0.923	0.244	0.244	1.000	0.416
Owner characteristics								
Age	27.8	27.2	27.5	0.320	27.0	27.0	0.938	0.141
Secondary Education	0.13	0.17	0.11	0.214	0.26	0.22	0.403	0.000
Certificate-Level Education	0.07	0.06	0.06	0.926	0.06	0.04	0.306	0.833
University and College Education	0.72	0.68	0.74	0.371	0.65	0.70	0.214	0.043
Postgraduate Education	0.09	0.10	0.09	0.905	0.03	0.03	1.000	0.005
Lived abroad at Baseline	0.03	0.01	0.04	0.270	0.00	0.02	0.171	0.024
Sector Ag, Farming and Fishery	0.20	0.21	0.21	0.982	0.13	0.20	0.088	0.056
Sector Wholesale and Retail	0.20	0.19	0.20	0.972	0.24	0.22	0.588	0.329
Sector ITC	0.10	0.14	0.11	0.641	0.10	0.08	0.536	0.742
Sector Manufacturing	0.05	0.11	0.13	0.047	0.11	0.10	0.793	0.652
Sector Prof, Scientific & Technical	0.08	0.10	0.08	0.695	0.10	0.07	0.229	0.341
Employment Income	155	173	219	0.140	138	144	0.790	0.012
Primary outcomes at baseline								
Has Business	0.748	0.749	0.746	0.997	0.648	0.704	0.182	0.020
Conditional on Having a Business:								
Has More than 1 Business	0.20	0.28	0.25	0.284	0.15	0.11	0.207	0.032
Sales	2456	2900	3317	0.522	1587	1408	0.614	0.012
Profits	601	439	577	0.562	437	400	0.774	0.329
Profits from All Businesses	641	575	652	0.901	449	412	0.779	0.194
Number of Employees	6.8	6.6	6.8	0.979	5.4	4.5	0.245	0.081
Percent of Paid Employees	0.788	0.87	0.87	0.126	0.90	0.79	0.042	0.521
Hours Worked in Business	43.134	39.553	41.353	0.511	38.062	35.619	0.426	0.319
Value of Inventory	4501	3679	2853	0.369	2267	2444	0.840	0.498
Value of Equipment	47230	44169	53436	0.474	21595	24417	0.455	0.000
Salary Paid to Self	181	152	226	0.035	154	145	0.745	0.018
Business Practices Index	0.684	0.694	0.698	0.880	0.752	0.708	0.028	0.029
Innovation & Training Index	0.499	0.521	0.521	0.498	0.465	0.447	0.436	0.014
Borrowed in Past from MFI Bank	0.222	0.278	0.220	0.363	0.241	0.170	0.110	0.648
Business Name Registered	0.645	0.671	0.699	0.565				
Omnibus Test of Joint Orthogonality (P-value)				0.361			0.568	

Notes: For age, information from intake is used to account for the fact that the two baselines did not take place at the same point of time. All monetary variables have been deflated to real Kenyan shillings using July 2024 prices, and have then converted to US-dollars using the July 2024 exchange rate. Sales and Profit variables are reported for the last month. Number of Employees, Sales, Value of Inventory, Employment Income, Salary, and hours worked have been winsorized at the 99 percentile. Value of equipment has been winsorized at the 95th percentile. Profits have been winsorized at the 1st and 99th percentiles.

Table 2: Impacts on Firm Operation and Employment

		1 Year Follow-up (1)	2 Year Follow-up (2)	3 Year Follow-up (3)	(1)=(2)=(3) (4)	Pooled (5)
Panel A: Has Business	36k	0.163*** (0.028)	0.196*** (0.039)	0.143*** (0.045)	0.644	0.166*** (0.025)
	9k-Multi	0.143*** (0.029)	0.190*** (0.039)	0.113** (0.046)	0.403	0.147*** (0.026)
	9k-Stream	0.223*** (0.032)	0.218*** (0.041)	0.079* (0.044)	0.018	0.172*** (0.027)
	Observations Multi	629	496	574		1699
	Observations Stream	440	362	408		1210
	Control Multi Mean	0.834	0.763	0.654		0.748
	Control Stream Mean	0.742	0.688	0.663		0.699
	P-value: 36k = 9k-Multi	0.156	0.800	0.473		0.260
	P-value: All Coefficients = 0	0.000	0.000	0.003		0.000
	P-value: 36k = 4*9k-Multi	0.000	0.000	0.054		0.000
	P-value: 9k-Multi = 9k-Stream	0.064	0.624	0.588		0.506
	36k	5.038*** (0.631)	3.364*** (0.669)	2.633*** (0.550)	0.014	3.726*** (0.458)
	9k-Multi	2.587*** (0.570)	2.103*** (0.623)	2.028*** (0.522)	0.744	2.239*** (0.421)
	9k-Stream	2.632*** (0.478)	1.591*** (0.423)	0.588 (0.364)	0.002	1.651*** (0.317)
Panel B: Number of Employees	Observations Multi	589	487	508		1584
	Observations Stream	419	362	372		1153
	Control Multi Mean	3.851	2.919	1.993		2.947
	Control Stream Mean	2.719	2.224	2.022		2.337
	P-value: 36k = 9k-Multi	0.000	0.043	0.345		0.003
	P-value: All Coefficients = 0	0.000	0.000	0.000		0.000
	P-value: 36k = 4*9k-Multi	0.013	0.022	0.006		0.001
	P-value: 9k-Multi = 9k-Stream	0.951	0.497	0.024		0.263
	36k	0.218*** (0.039)	0.114*** (0.034)	0.097*** (0.032)	0.040	0.144*** (0.025)
	9k-Multi	0.086** (0.036)	0.093*** (0.034)	0.086*** (0.031)	0.985	0.086*** (0.023)
Panel C: Has 10 or More Employees	9k-Stream	0.049* (0.027)	0.048* (0.026)	0.010 (0.023)	0.419	0.037** (0.017)
	Observations Multi	589	487	508		1584
	Observations Stream	419	362	372		1153
	Control Multi Mean	0.118	0.065	0.054		0.081
	Control Stream Mean	0.064	0.047	0.049		0.054
	P-value: 36k = 9k-Multi	0.001	0.570	0.762		0.037
	P-value: All Coefficients = 0	0.000	0.001	0.005		0.000
	P-value: 36k = 4*9k-Multi	0.339	0.042	0.036		0.024
	P-value: 9k-Multi = 9k-Stream	0.417	0.294	0.050		0.095

Notes:

Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. The pooled specification includes survey-wave fixed effects. Standard errors are robust and, in the pooled specification, clustered at the respondent level. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Column 4 reports p-values for tests of equality across the first, second, and third follow-up rounds. Row-level p-values test (i) whether the effect of receiving a 36k grant differs from that of receiving a 9k grant in the multi-staged competition; (ii) whether the effect of receiving a 36k grant is four times that of receiving a 9k grant in the multi-staged competition; (iii) whether the effects of all three grant types are jointly equal to zero; and (iv) whether the effect of receiving 9k in the multi-staged competition differs from that in the streamlined process. **Has Business** is an indicator equal to one if the respondent was operating a business at the time of the survey. **Number of Employees** includes all workers employed by the business, excluding the owner, and is winsorized at the 99th percentile. **Has 10 or More Employees** is an indicator equal to one if the business employs at least 10 workers.

Table 3: Impacts on Firm Performance

		1 Year Follow-up (1)	2 Year Follow-up (2)	3 Year Follow-up (3)	(1)=(2)=(3) (4)	Pooled (5)
Panel A: Monthly Sales	36k	1097*** (423)	892** (421)	953** (402)	0.938	982*** (316)
	9k-Multi	683 (453)	997** (485)	895* (465)	0.886	839** (360)
	9k-Stream	230 (303)	151 (313)	-184 (213)	0.455	79 (188)
	Observations Multi	606	488	507		1601
	Observations Stream	428	362	371		1161
	Control Multi Mean	1731	1330	1155		1424
	Control Stream Mean	1303	1078	941		1116
	P-value: 36k = 9k-Multi	0.362	0.821	0.905		0.701
	P-value: All Coefficients = 0	0.060	0.108	0.057		0.010
	P-value: 36k = 4*9k-Multi	0.325	0.081	0.132		0.078
	P-value: 9k-Multi = 9k-Stream	0.406	0.143	0.035		0.061
	36k	101 (128)	268** (117)	299*** (111)	0.465	210** (89)
	9k-Multi	96 (126)	259* (136)	225** (111)	0.629	183* (95)
	9k-Stream	-38 (108)	51 (80)	-77 (62)	0.443	-18 (63)
Panel B: Monthly Profits	Observations Multi	605	485	507		1597
	Observations Stream	427	361	371		1159
	Control Multi Mean	415	338	291		352
	Control Stream Mean	357	234	275		293
	P-value: 36k = 9k-Multi	0.966	0.942	0.534		0.774
	P-value: All Coefficients = 0	0.827	0.096	0.022		0.089
	P-value: 36k = 4*9k-Multi	0.528	0.119	0.142		0.135
	P-value: 9k-Multi = 9k-Stream	0.422	0.187	0.018		0.080
Panel C: Business Performance Index	36k	0.097 (0.092)	0.186 (0.115)	0.229* (0.117)	0.645	0.164** (0.082)
	9k-Multi	0.064 (0.100)	0.198 (0.135)	0.189 (0.130)	0.639	0.143 (0.094)
	9k-Stream	-0.070 (0.081)	-0.007 (0.097)	-0.120* (0.073)	0.641	-0.061 (0.060)
	Observations Multi	606	488	507		1601
	Observations Stream	428	362	371		1161
	Control Multi Mean	0.067	0.092	0.153		0.103
	Control Stream Mean	0.089	0.098	0.131		0.106
	P-value: 36k = 9k-Multi	0.737	0.923	0.765		0.826
	P-value: All Coefficients = 0	0.602	0.370	0.072		0.131
	P-value: 36k = 4*9k-Multi	0.660	0.218	0.275		0.240
	P-value: 9k-Multi = 9k-Stream	0.299	0.218	0.039		0.066

Notes:

Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. The pooled specification includes survey-wave fixed effects. Standard errors are robust and, in the pooled specification, clustered at the respondent level. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Column 4 reports p-values for tests of equality across the first, second, and third follow-up rounds. Row-level p-values test (i) whether the effect of receiving a 36k grant differs from that of receiving a 9k grant in the multi-staged competition; (ii) whether the effect of receiving a 36k grant is four times that of receiving a 9k grant in the multi-staged competition; (iii) whether the effects of all three grant types are jointly equal to zero; and (iv) whether the effect of receiving 9k in the multi-staged competition differs from that in the streamlined process. **Monthly Sales** are sales reported from the last month winsorized at the 99th percentile and deflated to real Kenyan shillings using 2024 prices, and then converted to US-dollars. **Monthly Profits** are winsorized at the 1st and 99th percentiles; all other transformations are the same. **Business Performance Index** is measured as an index averaging standardized (z-scored) monthly sales and profits. Each component is standardized using the control group mean and standard deviation, and the index is computed as the mean of these standardized variables.

Table 4: Robustness Checks for 3 Year Follow-up

		OLS	OLS with BL Controls	No Profits (are closed)	Personal Max	Average of One- Time Only	Lee Bound Treatment vs Control	Lee Bound 36k vs 9k-Multi
		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Number of Employees	36k	2.633*** (0.550)	2.541*** (0.570)	2.166*** (0.415)	2.217*** (0.553)	1.726*** (0.392)	[0.342, 3.472***] [(0.345), (0.587)]	[0.599, 0.749] [(0.642), (0.619)]
	9k-Multi	2.028*** (0.522)	2.111*** (0.533)	1.790*** (0.389)	1.044* (0.533)	1.353*** (0.371)	[-0.087, 2.770***] [(0.340), (0.559)]	
	9k-Stream	0.588 (0.364)	0.668* (0.361)	0.428 (0.292)	1.092*** (0.350)	0.399 (0.285)	[0.474, 0.654*] [(0.348), (0.365)]	
	Observations Multi	508	508	750	750	750	[449, 508]	[359, 359]
	Observations Stream	372	372	500	500	500	[370, 372]	
	Control Multi Mean	1.993	1.993	1.188	3.880	2.646	1.993	[5.428, 5.375]
	Control Stream Mean	2.022	2.022	1.496	2.000	2.434	2.022	
	P-value: 36k = 9k-Multi	0.345	0.486	0.451	0.048	0.426		
	P-value: All Coefficients = 0	0.000	0.000	0.001	0.321	0.011		
	P-value: 36k = 4*9k-Multi	0.006	0.003	0.000	0.000	0.000		
	P-value: 9k-Multi = 9k-Stream	0.024	0.025	0.005	0.940	0.042		
	36k	0.097*** (0.032)	0.088*** (0.030)	0.079*** (0.022)	0.089*** (0.029)	0.067*** (0.022)	[-0.052***, 0.125***] [(0.018), (0.035)]	[0.009, 0.014] [(0.036), (0.036)]
	9k-Multi	0.086*** (0.031)	0.093*** (0.031)	0.070*** (0.022)	0.059** (0.027)	0.059*** (0.021)	[-0.051***, 0.112***] [(0.018), (0.034)]	
	9k-Stream	0.010 (0.023)	0.020 (0.022)	0.008 (0.017)	0.032 (0.022)	0.007 (0.017)	[-0.000, 0.010] [(0.021), (0.021)]	
Panel B: Has 10 or More Employees	Observations Multi	508	508	750	750	750	[455, 508]	[359, 359]
	Observations Stream	372	372	500	500	500	[372, 372]	
	Control Multi Mean	0.054	0.054	0.032	0.116	0.070	0.054	0.096
	Control Stream Mean	0.049	0.049	0.036	0.052	0.060	0.049	
	P-value: 36k = 9k-Multi	0.762	0.891	0.749	0.355	0.751		
	P-value: All Coefficients = 0	0.005	0.004	0.016	0.147	0.040		
	P-value: 36k = 4*9k-Multi	0.036	0.016	0.000	0.004	0.003		
	P-value: 9k-Multi = 9k-Stream	0.050	0.054	0.025	0.436	0.060		
Panel C: Monthly Profits	36k	299*** (111)	300*** (112)	251*** (78)	234** (91)	204*** (76)	[-79, 378***] [(60), (122)]	[87, 157] [(117), (103)]
	9k-Multi	225** (111)	227** (113)	206*** (78)	110 (94)	160** (77)	[-123**, 308**] [(59), (122)]	
	9k-Stream	-77 (62)	-66 (55)	-60 (46)	-11 (81)	-62 (46)	[-75, -35] [(62), (47)]	
	Observations Multi	507	507	750	750	750	[448, 507]	[358, 358]
	Observations Stream	371	371	500	500	500	[371, 370]	
	Control Multi Mean	291	291	174	438	330	291	[525, 498]
	Control Stream Mean	275	275	203	307	304	[275, 239]	
	P-value: 36k = 9k-Multi	0.534	0.499	0.619	0.187	0.611		
	P-value: All Coefficients = 0	0.022	0.028	0.052	0.548	0.134		
	P-value: 36k = 4*9k-Multi	0.142	0.133	0.002	0.084	0.015		
	P-value: 9k-Multi = 9k-Stream	0.018	0.020	0.003	0.330	0.013		

Notes:
Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. The pooled specification includes survey-wave fixed effects. Standard errors are robust and, in the pooled specification, clustered at the respondent level. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Row-level p-values test (i) whether the effect of receiving a 36k grant differs from that of receiving a 9k grant in the multi-staged competition; (ii) whether the effect of receiving a 36k grant is four times that of receiving a 9k grant in the multi-staged competition; (iii) whether the effects of all three grant types are jointly equal to zero; and (iv) whether the effect of receiving 9k in the multi-staged competition differs from that in the streamlined process. **Number of Employees** includes all workers employed by the business, excluding the owner, and is winsorized at the 99th percentile. **Has 10 or More Employees** is an indicator equal to one if the business employs at least 10 workers. **Monthly Profits** are winsorized at the 1st and 99th percentiles and deflated to real Kenyan shillings using 2024 prices, and then converted to US-dollars. **Column 1** shows the base specification. **Column 2** adds the variables summarized in Table 1 as controls. **Column 3** assumes that all attritors are closed and therefore have zero employees and zero profits. **Column 4** assigns attritors the maximum value observed for that respondent across the baseline and three follow-up rounds. **Column 5** assigns attritors the average outcome among respondents observed in only one of the three follow-up surveys. **Columns 6 and 7** report Lee (2009) bounds, with upper and lower bounds shown in parentheses. Column 6 presents bounds for the comparison between the control and treatment groups, while Column 7 presents bounds for the comparison between the 36k and 9k multi-stage treatment arms.

Table 5: Heterogeneity by Gender

	Number of Employees (1)	More Than 10 Employees (2)	Profits (3)
36k	3.592*** (0.543)	0.143*** (0.030)	165* (93)
36k * Female	0.532 (0.996) [1.000]	0.004 (0.054) [1.000]	192 (249) [1.000]
9k-Multi	2.380*** (0.517)	0.093*** (0.029)	208* (109)
9k-Multi * Female	-0.624 (0.789) [1.000]	-0.033 (0.041) [1.000]	-106 (229) [1.000]
9k-Stream	1.375*** (0.512)	0.023 (0.028)	-61 (103)
9k-Stream * Female	0.593 (0.617) 1.000	0.030 (0.034) 1.000	92 (125) 1.000
Female Dummy Multi	0.327 (0.615)	-0.003 (0.033)	-183 (206)
Female Dummy Stream	-1.208*** (0.428)	-0.066*** (0.023)	-144 (91)
Number Obs Multi	1584	1584	1597
Number Obs Stream	1153	1153	1159
Control Multi Female Mean	1.533	0.011	289
Control Multi Male Mean	3.328	0.100	369
Control Stream Female Mean	1.531	0.023	170
Control Stream Male Mean	3.020	0.079	396
P-val: 36k * Female = 9k-Multi * Female	0.295	0.540	0.151
P-val: 9k-Multi * Female = 9k-Stream * Female	0.220	0.244	0.448

Notes:

This table reports heterogeneity in treatment effects by gender, interacting each treatment indicator with a dummy equal to one if the respondent is female. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Column (1) reports results for **Number of Employees**, which includes all workers employed by the business, excluding the owner, and is winsorized at the 99th percentile. Column (2) for **Has 10 or More Employees**, an indicator equal to one if the business employs at least 10 workers and Column (3) for **Monthly Profits**, winsorized at the 1st and 99th percentiles and deflated to real Kenyan shillings using 2024 prices, and then converted to US-dollars. All results are estimated using the pooled specification. Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values and survey-wave fixed effects. Standard errors are robust and clustered at the respondent level.

Sharpened q-values (Benjamini, Krieger, and Yekutieli (2006)) for the interaction terms are presented in square brackets and are computed jointly across all 36 heterogeneity tests (4 heterogeneity dimensions × 3 treatment arms × 3 outcomes). The heterogeneity dimensions are gender, business ownership at intake, city, and education.

Table 6: Heterogeneity by Business Ownership at Intake

	Number of Employees (1)	More than 10 employees (2)	Profits (3)
36k	3.498*** (0.629)	0.113*** (0.032)	21 (103)
36k * Have Biz	0.363 (0.883)	0.050 (0.048)	307* (165)
	1.000	1.000	1.000
9k-Multi	2.321*** (0.645)	0.082** (0.036)	99 (120)
9k-Multi * Have Biz	-0.134 (0.842)	0.006 (0.047)	133 (179)
	1.000	1.000	1.000
9k-Stream	1.902*** (0.333)	0.039** (0.020)	-25 (57)
9k-Stream * Have Biz	-0.471 (0.625)	-0.004 (0.034)	15 (124)
	1.000	1.000	1.000
Have Biz Dummy Multi	0.522 (0.704)	-0.012 (0.038)	-203.438 (156.626)
Have Biz Dummy Stream	0.978* (0.582)	0.021 (0.030)	173.511 (132)
Number Obs Multi	1584	1584	1597
Number Obs Stream	1153	1153	1159
Control Multi Have Biz Mean	3.593	0.103	405
Control Multi No Biz Mean	1.844	0.044	260
Control Stream Have Biz Mean	3.311	0.078	411
Control Stream No Biz Mean	1.237	0.027	160
P-val: 36k * Have Biz = 9k-Multi * Have Biz	0.610	0.432	0.318
P-val: 9k-Multi * Have Biz = 9k-Stream * Have Biz	0.748	0.866	0.590

Notes:

This table reports heterogeneity in treatment effects by business ownership at intake, interacting each treatment indicator with a dummy equal to one if the respondent is owned a businesses at time of applying. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Column (1) reports results for **Number of Employees**, which includes all workers employed by the business, excluding the owner, and is winsorized at the 99th percentile. Column (2) for **Has 10 or More Employees**, an indicator equal to one if the business employs at least 10 workers and Column (3) for **Monthly Profits**, winsorized at the 1st and 99th percentiles and deflated to real Kenyan shillings using 2024 prices, and then converted to US-dollars. All results are estimated using the pooled specification. Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values and survey-wave fixed effects. Standard errors are robust and clustered at the respondent level.

Sharpened q-values (Benjamini, Krieger, and Yekutieli (2006)) for the interaction terms are presented in square brackets and are computed jointly across all 36 heterogeneity tests (4 heterogeneity dimensions × 3 treatment arms × 3 outcomes). The heterogeneity dimensions are gender, business ownership at intake, city, and education.

Table 7: Impact of Grant Use on Capital and Finance

	Land and Buildings		Inventory Value		Capital Stock		Took out a Loan	Total amount of Loans Taken
	1 Year Follow-up (1)	3 Year Follow-up (2)	1 Year Follow-up (3)	3 Year Follow-up (4)	1 Year Follow-up (5)	3 Year Follow-up (6)	Combined Across the Three Follow-up Rounds (7)	Combined Across the Three Follow-up Rounds (8)
36k	28378*** (7287)	3084** (1337)	4759*** (904)	1588.544* (862)	7271** (2856)	4990*** (1454)	0.022 (0.044)	1672** (799.754)
9k-Multi	23885*** (7810)	1177 (1212)	4193*** (940)	1607* (893)	4291 (3366)	2365 (1461)	0.053 (0.044)	214 (666.332)
9k-Stream	19486*** (6209)	214 (846)	1802*** (600)	227 (427)	6434** (2732)	1051 (1125)	0.016 (0.044)	132 (649.588)
Observations Multi	551	421	589	425	629	574	634	654
Observations Stream	394	320	418	325	440	408	449	463
Control Multi Mean	18676	2100	2020	1462	12544	3077	0.291	1757
Control Stream Mean	12999	1618	1215	966	7345	3230	0.323	1866
P-value: 36k = 9k-Multi	0.616	0.159	0.611	0.984	0.345	0.108	0.470	0.073
P-value: All Coefficients = 0	0.000	0.146	0.000	0.187	0.008	0.005	0.663	0.195
P-value: 36k = 4*9k-Multi	0.025	0.714	0.001	0.139	0.421	0.414	0.231	0.740
P-value: 9k-Multi = 9k-Stream	0.659	0.515	0.032	0.164	0.621	0.476	0.548	0.929

Notes:

Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. Standard errors are robust. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. **Land and buildings**, **Inventory value**, and **Capital stock** are winsorized at the 99th percentile. Capital stock excludes land and buildings as well as inventory. These variables are reported for the first and third follow-up surveys; data were not collected in the second follow-up. **Took out a loan** indicator equals one if the respondent reports taking a loan in any follow-up round. **Amount of loans** variable is the sum of loans reported across all follow-up rounds and is winsorized at the 1st and 99th percentiles. Row-level p-values test (i) whether the effect of receiving a 36k grant differs from that of receiving a 9k grant in the multi-stage competition; (ii) whether the effect of receiving a 36k grant is four times that of receiving a 9k grant in the multi-stage competition; (iii) whether the effects of all three grant types are jointly equal to zero; and (iv) whether the effect of receiving 9k in the multi-stage competition differs from that in the streamlined process.

Table 8: Impact of Grant on Business Organization and Innovation

	Business Practice Index		Has a Website		Innovation and Training Index	
	Pooled, unconditional (1)	Pooled, conditional (2)	3 Year Follow-up, unconditional (3)	3 Year Follow-up, conditional (4)	Pooled, unconditional (5)	Pooled, conditional (6)
36k	0.218*** (0.024)	0.079*** (0.015)	0.180*** (0.045)	0.188*** (0.069)	0.175*** (0.019)	0.101*** (0.017)
9k-Multi	0.181*** (0.024)	0.054*** (0.016)	0.162*** (0.044)	0.171** (0.068)	0.125*** (0.019)	0.055*** (0.017)
9k-Stream	0.151*** (0.025)	0.011 (0.015)	0.060* (0.036)	0.061 (0.057)	0.114*** (0.018)	0.049*** (0.017)
Observations Multi	1496	1259	427	277	1498	1261
Observations Stream	1102	837	325	201	1103	838
Control Multi Mean	0.498	0.721	0.092	0.188	0.299	0.432
Control Stream Mean	0.494	0.747	0.108	0.193	0.270	0.407
P-value: 36k = 9k Multi	0.045	0.048	0.708	0.804	0.004	0.004
P-value: All Coefficients = 0	0.000	0.000	0.000	0.019	0.000	0.000
P-value: 36k = 4*9k-Multi	0.000	0.013	0.005	0.041	0.000	0.051
P-value: 9k-Multi = 9k-Stream	0.389	0.047	0.075	0.214	0.667	0.809

Notes:

Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. Pooled specifications include survey-wave fixed effects. Standard errors are robust and, in pooled specifications, clustered at the respondent level.

The **Business Practice Index** follows David McKenzie and Christopher Woodruff (2017) and is defined as the proportion of recommended practices adopted. **Has a Website** is an indicator equal to one if the respondent reports having a website; this outcome is measured only at the 3-year follow-up, when the question was introduced. The **Innovation and Training Index** is the average of different measures of innovation and training activities listed in Appendix D. Results are reported both unconditionally and conditional on operating a business.

Row-level p-values test: (i) equality of the 36k and 9k grant effects within the multi-stage competition; (ii) whether the 36k effect equals four times the 9k effect in the multi-stage competition; (iii) whether all treatment effects are jointly equal to zero; and (iv) whether the 9k effect differs between the multi-stage competition and the streamlined process.

Table 9: Reweighting the Streamlined Sample to Look More Similar to the Multi-Stage Sample on Gender, New Vs Existing, and Application Score

		Have Business (1)	Number of Employees (2)	More than 10 employees (3)	Profits (4)	Sales (5)	Z-score Business Performance (6)
Panel A: Reweighting the Streamlined Sample to have same Gender and New/Existing Proportions as Multi-Stage Sample	36k	0.167*** (0.025)	3.727*** (0.458)	0.149*** (0.026)	210** (89)	982*** (316)	0.164** (0.082)
	9k-Multi	0.147*** (0.026)	2.240*** (0.421)	0.087*** (0.023)	183* (95)	839** (360)	0.143 (0.094)
	9k-Stream	0.153*** (0.028)	1.465*** (0.438)	0.028 (0.024)	-36 (90)	29 (245)	-0.071 (0.082)
	Observations Multi	1699	1584	1584	1597	1601	1601
	Observations Stream	1210	1153	1153	1159	1161	1161
	Control Multi Mean	0.748	2.947	0.081	352	1424	0.103
	Control Stream Mean	0.699	2.337	0.054	293	1116	0.106
	P-value: 9k-Multi = 9k-Stream reweighted	0.878	0.202	0.082	0.095	0.063	0.086
	36k	0.168*** (0.025)	3.718*** (0.457)	0.144*** (0.025)	210** (89)	982*** (316)	0.163** (0.082)
	9k-Multi	0.148*** (0.026)	2.234*** (0.420)	0.086*** (0.023)	183* (95)	840** (360)	0.143 (0.094)
Panel B: IPW Reweighted Results Based on First-Stage Application Score	9k-Stream	0.134*** (0.047)	0.910 (0.582)	-0.019 (0.032)	-384 (334)	-920 (1080)	-0.355 (0.291)
	Observations Multi	1699	1584	1584	1597	1601	1601
	Observations Stream	1210	1153	1153	1159	1161	1161
	Control Multi Mean	0.748	2.947	0.081	352	1424	0.103
	Control Stream Mean	0.716	3.020	0.102	639	2175	0.400
	P-value: 9k-Multi = 9k-Stream reweighted	0.786	0.064	0.008	0.104	0.123	0.105

Notes:

Panel A reweights observations in the streamlined sample so that the joint distribution of gender and business status (new vs. existing) matches that of the multi-stage sample. Specifically, weights are constructed as the ratio of the share of each gender × business-status stratum in the multi-stage sample (see Table 1) to its share in the streamlined sample. Observations in the multi-stage sample are assigned weight one.

Panel B implements inverse probability weighting (IPW) based on first-stage application scores. We estimate weights so that the distribution of first-stage scores in the streamlined sample matches that of the multi-stage sample, and apply these weights to the streamlined observations; multi-stage observations again receive weight one.

All regressions are estimated pooling data from the three follow-up surveys, using these weights, include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline and survey-wave fixed effects. Standard errors are robust and clustered at the respondent level. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Column 4 reports p-values for tests of equality across the first, second, and third follow-up rounds. **Has Business** is an indicator equal to one if the respondent was operating a business at the time of the survey. **Number of Employees** includes all workers employed by the business, excluding the owner, and is winsorized at the 99th percentile. **Has 10 or More Employees** is an indicator equal to one if the business employs at least 10 workers. **Monthly Profits** are winsorized at the 1st and 99th percentiles; all other transformations are the same. **Monthly Sales** are sales reported from the last month winsorized at the 99th percentile and deflated to real Kenyan shillings using 2024 prices, and then converted to US-dollars. **Business Performance Index** is measured as an index averaging standardized (z-scored) monthly sales and profits. Each component is standardized using the control group mean and standard deviation, and the index is computed as the mean of these standardized variables.

Figures

Figure 1: Impact Evaluation Process

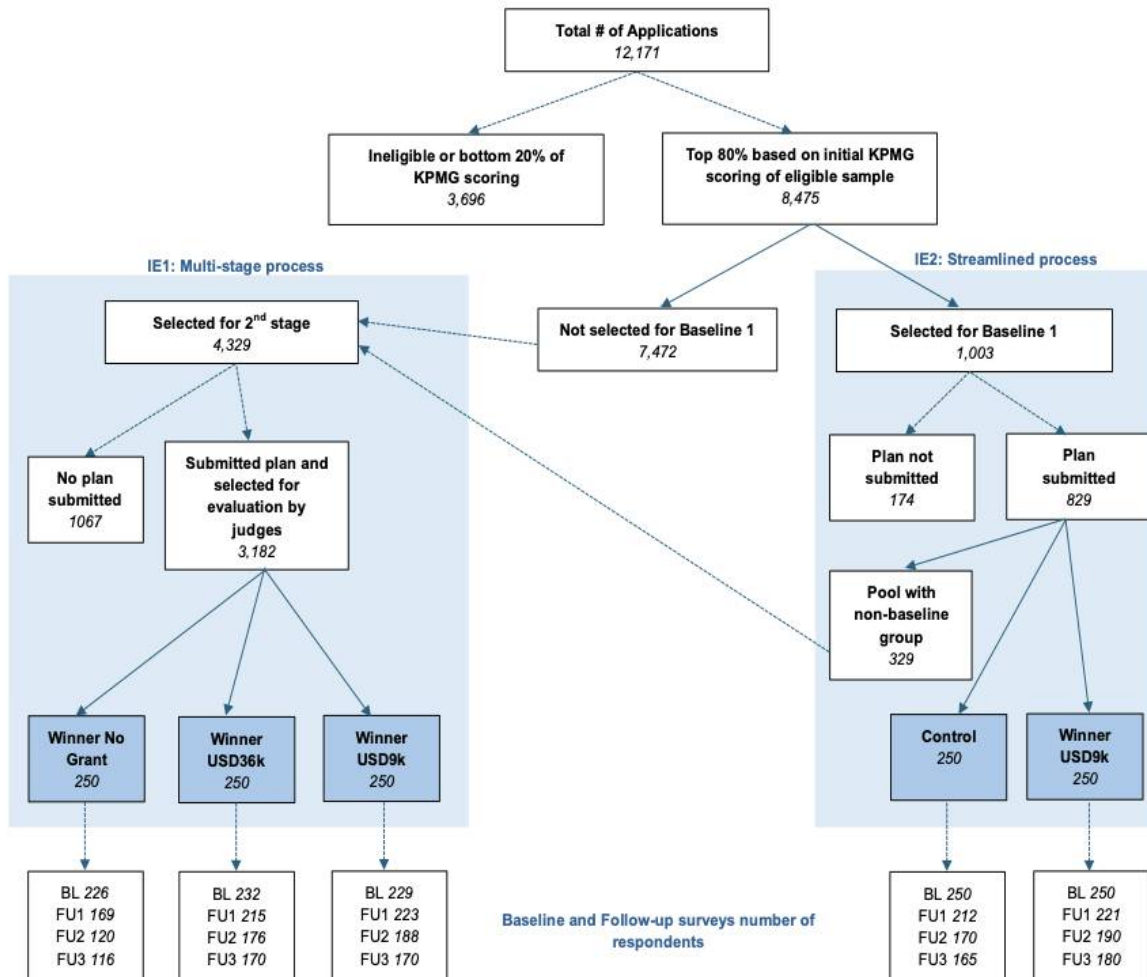
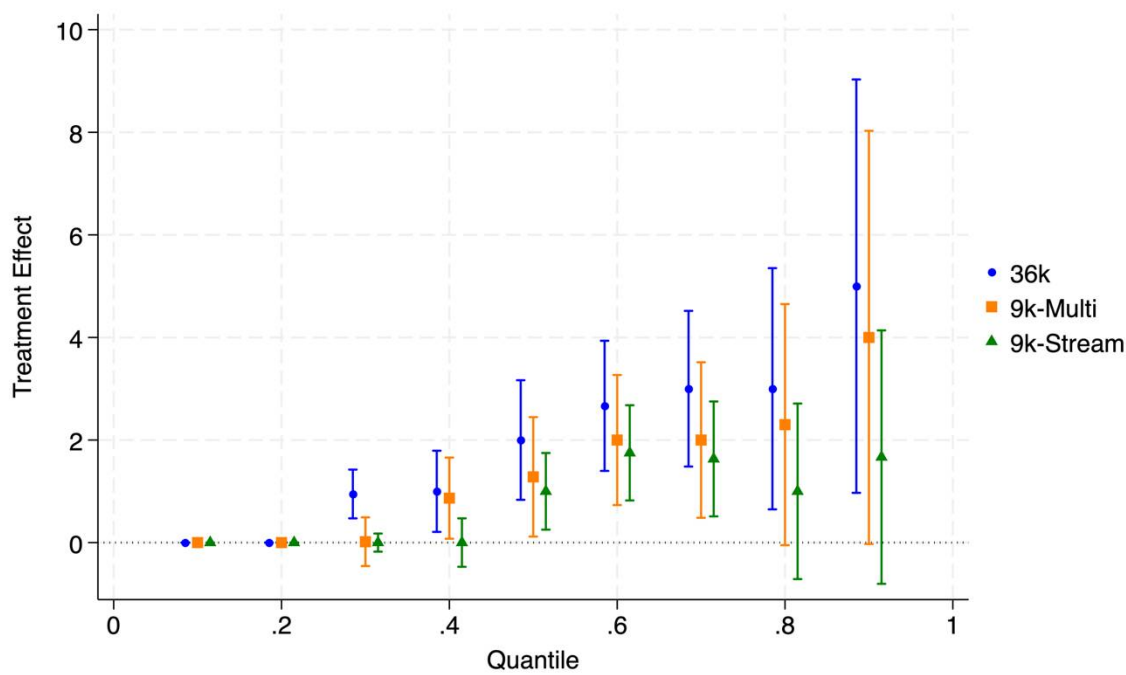


Figure 2: Quantile Treatment Effects (3-year follow-up survey)

Panel A: Number of Employees



Panel B: Profits

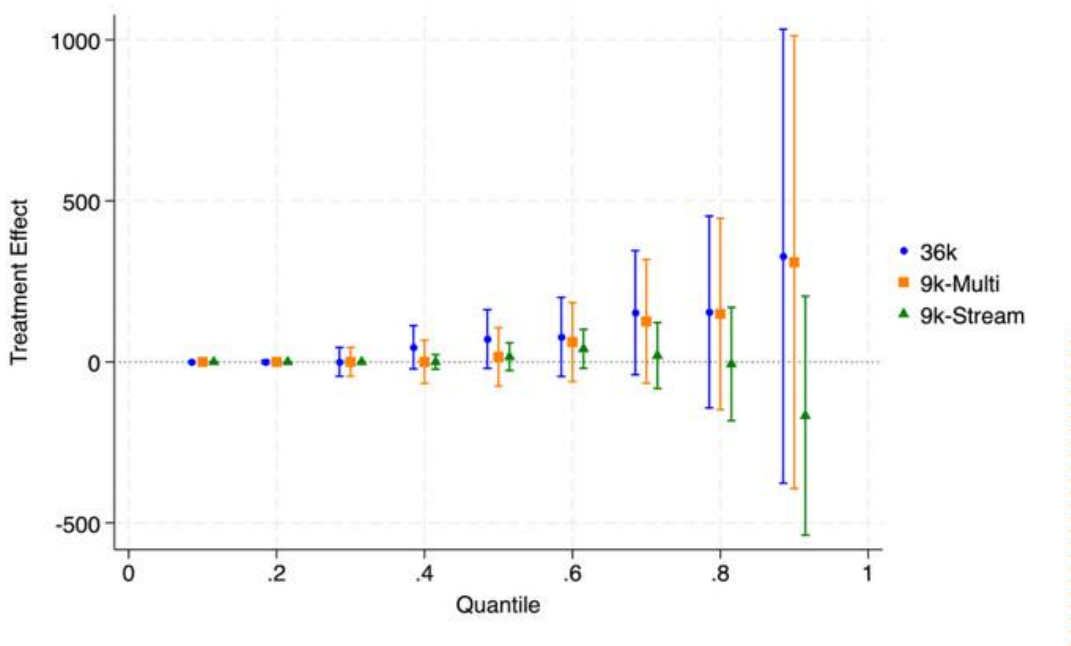


Figure 3: Grant Expenditure by Treatment Group

Appendices

A. Timeline

Applications received	June-September 2019
Baseline for Streamlined Sample	February 2020
Business Plan Training	July-August 2020
Business plans due	August 31, 2020
Business Plan Screening and Selection	January 2021
Baseline for Multi-stage sample	January-February 2021
Award Ceremony	February 2021
Provision of Grants and Monitoring	May 2021-May 2022
First follow-up survey	May-August 2022
Second follow-up survey	February-May 2023
Third follow-up survey	July 2024-May 2025

Table A1: Comparison of Winners in Both Processes to the Other Applicants

	Winners Multi- stage	Winners Streamlined	Other Applicants	P-Value: Multi-Stage vs Other Applicants	P-value: Streamlined vs Other Applicants
Personal Characteristics					
Age	27.5	27.0	26.4	0.000	0.002
Female	0.26	0.49	0.22	0.027	0.000
City: Nairobi/Mombasa/Kisumu	0.37	0.33	0.32	0.018	0.679
Education = Secondary or Less	0.13	0.18	0.24	0.000	0.004
Education = Higher than Secondary	0.87	0.82	0.76	0.000	0.004
Business Owner (Employer and Self-Employed)	0.53	0.48	0.45	0.000	0.299
Wage Employed	0.18	0.16	0.15	0.023	0.595
Student	0.14	0.17	0.17	0.050	0.738
Unemployed	0.14	0.20	0.23	0.000	0.167
Business Characteristics					
Has Business	0.60	0.52	0.47	0.000	0.026
Business Registered	0.81	0.76	0.65	0.000	0.000
Number of Employees	10.3	4.4	6.4	0.261	0.617
Sector = Agriculture, Farming and Fishery	0.29	0.23	0.23	0.000	0.710
Sector = Wholesale and Retail Trade	0.05	0.05	0.08	0.000	0.003
Sector = Information And Communication	0.07	0.07	0.10	0.005	0.079
Sector = Manufacturing	0.12	0.08	0.06	0.000	0.067
Sector = Arts, Entertainment And Recreation	0.04	0.07	0.08	0.000	0.596
Business Age	2.2	2.2	2.2	0.780	0.862
Streamlined Competition Score	37.57	31.38	25.06	0.000	0.000
Number of Observations	750	500	10216		

Notes:

This Table used intake information collected in 2019. The highest education level attained is collected with less nuance than at baseline (Table 1); thus "Higher than Secondary" combines the categories "Certificate level" and "University and College". Sector is reported unconditionally, taking into account not only existing businesses but also planned businesses. For 7 winners (2 from the multi-stage and 5 from streamlined competition) we were not able to match them with the intake data and hence they are missing in this table.

Table A2: Secondary employment outcomes

		1 Year Follow-up (1)	2 Year Follow-up (2)	3 Year Follow-up (3)	(1)=(2)=(3) (4)	Pooled (5)
More than One Business	36k	0.080* (0.043)	0.078 (0.051)	0.103*** (0.039)	0.891	0.087*** (0.029)
	9k-Multi	0.069* (0.042)	0.027 (0.051)	0.035 (0.036)	0.767	0.046 (0.029)
	9k-Stream	0.158*** (0.036)	0.082* (0.043)	0.041 (0.032)	0.048	0.097*** (0.025)
	Observations Multi	629	488	508		1625
	Observations Stream	440	362	372		1174
	Control Multi Mean	0.221	0.260	0.101		0.192
	Control Stream Mean	0.111	0.188	0.092		0.128
	P-value: 36k = 9k-Multi	0.798	0.285	0.085		0.142
	P-value: All Coefficients = 0	0.000	0.105	0.035		0.000
	P-value: 36k = 4*9k-Multi	0.193	0.859	0.780		0.350
	P-value: 9k-Multi = 9k-Stream	0.108	0.416	0.908		0.177
Hours Worked	36k	14.376*** (2.612)	12.129*** (2.587)	6.022** (2.780)	0.076	11.171*** (1.856)
	9k-Multi	9.480*** (2.606)	7.017*** (2.483)	4.494* (2.696)	0.408	7.144*** (1.753)
	9k-Stream	9.681*** (2.655)	7.988*** (2.509)	8.463*** (2.496)	0.892	8.780*** (1.884)
	Observations Multi	609	488	508		1605
	Observations Stream	433	362	372		1167
	Control Multi Mean	28.781	25.553	22.980		25.921
	Control Stream Mean	28.596	25.376	20.995		25.157
	P-value: 36k = 9k-Multi	0.053	0.030	0.540		0.017
	P-value: All Coefficients = 0	0.000	0.000	0.001		0.000
	P-value: 36k = 4*9k-Multi	0.012	0.068	0.205		0.005
	P-value: 9k-Multi = 9k-Stream	0.957	0.783	0.280		0.525
Number of Wage Workers	36k	2.308*** (0.425)	1.970*** (0.358)	2.153*** (0.809)	0.828	2.120*** (0.400)
	9k-Multi	0.895** (0.391)	2.064*** (0.676)	1.475*** (0.445)	0.280	1.425*** (0.360)
	9k-Stream	1.361*** (0.227)	0.917*** (0.236)	0.127 (0.283)	0.003	0.832*** (0.171)
	Observations Multi	586	481	508		1575
	Observations Stream	419	359	372		1150
	Control Multi Mean	1.807	1.050	1.007		1.318
	Control Stream Mean	1.034	0.918	1.130		1.030
	P-value: 36k = 9k-Multi	0.001	0.894	0.448		0.169
	P-value: All Coefficients = 0	0.000	0.000	0.001		0.000
	P-value: 36k = 4*9k-Multi	0.376	0.018	0.049		0.013
	P-value: 9k-Multi = 9k-Stream	0.302	0.110	0.011		0.137
Number of Casual Workers	36k	2.365*** (0.483)	-0.052 (1.994)	0.904*** (0.332)	0.033	1.175* (0.641)
	9k-Multi	1.805*** (0.409)	-1.107 (1.919)	0.984** (0.439)	0.164	0.680 (0.634)
	9k-Stream	0.850** (0.379)	0.384 (0.346)	0.563*** (0.211)	0.655	0.601*** (0.232)
	Observations Multi	588	481	508		1577
	Observations Stream	419	359	372		1150
	Control Multi Mean	1.112	2.975	0.711		1.497
	Control Stream Mean	1.143	1.035	0.557		0.916

	P-value: 36k = 9k-Multi	0.300	0.227	0.866	0.235
	P-value: 36k = 4*9k-Multi	0.000	0.414	0.001	0.016
	P-value: All coefficients = 0	0.002	0.462	0.074	0.457
	P-value: 9k-Multi = 9k-Stream	0.088	0.445	0.389	0.907
	36k	0.909** (0.436)	0.371** (0.153)	0.154 (0.103)	0.146 (0.212)
	9k-Multi	-0.071 (0.237)	0.007 (0.100)	-0.013 (0.071)	0.951 (0.090)
	9k-Stream	0.342** (0.138)	0.248** (0.105)	-0.135 (0.090)	0.002 (0.071)
Number of Commissioned Workers	Observations Multi	586	481	508	1575
	Observations Stream	419	359	372	1150
	Control Multi Mean	0.404	0.198	0.154	0.260
	Control Stream Mean	0.266	0.165	0.270	0.237
	P-value: 36k = 9k-Multi	0.051	0.004	0.103	0.013
	P-value: All Coefficients = 0	0.015	0.003	0.167	0.010
	P-value: 36k = 4*9k-Multi	0.257	0.305	0.438	0.154
	P-value: 9k-Multi = 9k-Stream	0.132	0.097	0.289	0.159
	36k	0.031 (0.051)	-0.003 (0.024)	-0.025 (0.041)	0.680 (0.026)
	9k-Multi	0.005 (0.052)	0.031 (0.034)	0.010 (0.047)	0.887 (0.030)
	9k-Stream	0.032 (0.038)	0.040 (0.029)	0.010 (0.013)	0.573 (0.018)
Number of Unpaid Workers	Observations Multi	587	481	508	1576
	Observations Stream	419	359	372	1150
	Control Multi Mean	0.118	0.041	0.047	0.072
	Control Stream Mean	0.074	0.018	0.011	0.036
	P-value: 36k = 9k-Multi	0.606	0.240	0.245	0.638
	P-value: All Coefficients = 0	0.765	0.344	0.556	0.454
	P-value: 36k = 4*9k-Multi	0.958	0.297	0.673	0.579
	P-value: 9k-Multi = 9k-Stream	0.684	0.848	0.990	0.742

Notes:

Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. The pooled specification includes survey-wave fixed effects. Standard errors are robust and, in the pooled specification, clustered at the respondent level. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Column 4 reports p-values for tests of equality across the first, second, and third follow-up rounds. Row-level p-values test (i) whether the effect of receiving a 36k grant differs from that of receiving a 9k grant in the multi-stage competition; (ii) whether the effect of receiving a 36k grant is four times that of receiving a 9k grant in the multi-stage competition; (iii) whether the effects of all three grant types are jointly equal to zero; and (iv) whether the effect of receiving 9k in the multi-stage competition differs from that in the streamlined process.

More than One Business is an indicator equal to one if the respondent operates more than one business. **Hours Worked** is the number of hours the entrepreneur spent working in the business in a typical week during the last month, winsorized at the 1st and 99th percentiles. **Number of Wage Workers** is the number of wage or salaried workers employed in the business at the time of the interview. **Number of Casual Workers** is the number of casual or daily-paid workers employed in the business at the time of the interview. **Number of Commissioned Workers** is the number of workers paid on commission at the time of the survey. **Number of Unpaid Workers** is the number of unpaid workers in the business at the time of the survey.

Table A3: 1 and 2 Year Follow-up Survey Highest Paid Workers Characteristics

	36k (1)	9k- Multi (2)	Control- Multi (3)	P-value (36k=9k- Multi=Control- Multi) (4)	p(36k=9k- Multi) (5)	9k- Stream (6)	Control- Stream (7)	P-value (9k- Stream=Control- Stream) (8)	P-value (9k- Stream=9k- Multi) (9)
1 Year Follow-up - Worker 1									
Age	29.3	29.8	30.6	0.226	0.392	29.1	29.9	0.368	0.316
Female	0.30	0.29	0.27	0.822	0.758	0.34	0.33	0.870	0.330
Related to Respondent	0.08	0.14	0.13	0.086	0.034	0.16	0.14	0.764	0.728
Found Job through Advertisement	0.17	0.11	0.06	0.011	0.071	0.09	0.03	0.019	0.528
Found Job through Family/Friends Network	0.35	0.42	0.53	0.019	0.201	0.4	0.5	0.477	0.759
Casual Worker	0.2	0.2	0.3	0.214	0.345	0.3	0.4	0.039	0.427
Wage Worker	0.8	0.7	0.6	0.003	0.202	0.7	0.6	0.008	0.510
Unpaid Worker	0.0	0.0	0.1	0.000	0.141	0.0	0.1	0.074	0.324
Highest Education Incomplete Secondary	0.06	0.12	0.14	0.050	0.062	0.06	0.11	0.138	0.057
Highest Education Secondary	0.32	0.32	0.35	0.871	0.873	0.35	0.35	0.991	0.453
Highest Education Incomplete Post-Secondary	0.02	0.02	0.02	0.825	0.537	0.03	0.03	0.970	0.463
Highest Education Completed Post-Secondary	0.59	0.55	0.50	0.270	0.425	0.56	0.51	0.420	0.865
Wage per Month	222	180	196	0.423	0.193	161	206	0.202	0.364
1 Year Follow-up - Worker 2									
Age	28.9	29.0	28.5	0.847	0.904	27.5	28.6	0.136	0.030
Female	0.40	0.47	0.46	0.331	0.160	0.48	0.33	0.019	0.903
Related to Respondent	0.06	0.12	0.04	0.047	0.040	0.09	0.07	0.670	0.434
Found Job through Advertisement	0.15	0.14	0.11	0.595	0.618	0.09	0.12	0.465	0.176
Found Job through Family/Friends Network	0.36	0.41	0.51	0.074	0.351	0.4	0.4	0.480	0.901
Casual Worker	0.2	0.3	0.3	0.212	0.101	0.3	0.5	0.011	0.666
Wage Worker	0.7	0.6	0.5	0.004	0.037	0.7	0.4	0.000	0.766
Unpaid Worker	0.0	0.0	0.1	0.001	0.180	0.0	0.1	0.022	0.107
Highest Education Incomplete Secondary	0.07	0.13	0.16	0.034	0.051	0.10	0.14	0.296	0.292
Highest Education Secondary	0.36	0.30	0.39	0.243	0.183	0.39	0.41	0.686	0.069
Highest Education Incomplete Post-Secondary	0.02	0.04	0.01	0.268	0.266	0.04	0.00	0.063	0.704
Highest Education Completed Post-Secondary	0.54	0.53	0.44	0.241	0.738	0.48	0.45	0.618	0.367
Wage per Month	168	168	198	0.695	0.989	144	164	0.543	0.373
2 Year Follow-up - Worker 1									
Age	30.0	29.5	30.2	0.756	0.536	28.4	29.0	0.440	0.141
Female	0.33	0.41	0.38	0.339	0.143	0.45	0.37	0.255	0.509
Related to Respondent	0.06	0.14	0.15	0.040	0.022	0.11	0.13	0.726	0.487
Found Job through Advertisement	0.21	0.10	0.11	0.019	0.008	0.09	0.05	0.334	0.759
Found Job through Family/Friends Network	0.34	0.41	0.31	0.319	0.216	0.4	0.5	0.618	0.746
Casual Worker	0.2	0.2	0.4	0.034	0.457	0.3	0.4	0.259	0.448
Wage Worker	0.8	0.7	0.6	0.021	0.451	0.7	0.6	0.046	0.517
Unpaid Worker	0.0	0.0	0.0	0.753	0.929	0.0	0.1	0.030	0.761
Highest Education Incomplete Secondary	0.05	0.10	0.17	0.019	0.103	0.06	0.08	0.554	0.200
Highest Education Secondary	0.29	0.34	0.30	0.704	0.414	0.39	0.38	0.869	0.356
Highest Education Incomplete Post-Secondary	0.03	0.00	0.00	0.075	0.035	0.01	0.04	0.129	0.287
Highest Education Completed Post-Secondary	0.63	0.56	0.53	0.293	0.214	0.54	0.50	0.577	0.745
Wage per Month	197	181	176	0.719	0.486	157	154	0.928	0.296

Notes:

The 1 year follow-up survey includes questions about the two employees with the highest wages you hired and who are still working with the entrepreneur. The 2 year follow-up survey asks mostly similar questions about the highest paid employee.

Table A4: Innovation and Training Index Components

	Introduction New Products	Characteristics New Products and Services	New Product/Service has been Copied by Competitor in City	Improved an Existing Product/Service in past 12 Months	Introduced New or Improved Business Processes in past 12 Months	Opened New Shop or Production Location in past 12 Months	Trained Staff in new Areas for the Business	Licensed a new Technology from Another Firm	Brought in- house Production that was Previously Outsourced	Introduced a New in- House Training Program
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
36k-Multi	0.261*** (0.033)	0.106*** (0.017)	0.113*** (0.027)	0.229*** (0.033)	0.210*** (0.033)	0.106*** (0.024)	0.283*** (0.034)	0.031* (0.019)	0.117*** (0.029)	0.158*** (0.030)
9k-Multi	0.205*** (0.034)	0.100*** (0.017)	0.048* (0.027)	0.216*** (0.034)	0.150*** (0.035)	0.097*** (0.023)	0.209*** (0.035)	0.005 (0.018)	0.075*** (0.027)	0.096*** (0.030)
9k-Stream	0.144*** (0.032)	0.076*** (0.017)	0.044 (0.028)	0.136*** (0.034)	0.119*** (0.030)	0.054*** (0.020)	0.158*** (0.031)	0.001 (0.014)	0.056** (0.022)	0.072*** (0.025)
Observations Multi	1491	1491	1359	1492	1498	1498	1498	1498	1498	1497
Observations Stream	1101	1101	1010	1098	1103	1103	1103	1103	1103	1103
Control Multi Mean	0.275	0.070	0.142	0.392	0.333	0.087	0.301	0.070	0.141	0.190
Control Stream Mean	0.321	0.089	0.175	0.405	0.299	0.079	0.252	0.051	0.098	0.153
P-value: 36k = 9k-Multi	0.090	0.755	0.020	0.682	0.077	0.723	0.027	0.153	0.147	0.045
P-value: All Coefficients = 0	0.000	0.000	0.421	0.000	0.002	0.001	0.000	0.859	0.064	0.039
P-value: 36k = 4*9k-Multi	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.366	0.000	0.000
P-value: 9k-Multi = 9k-Stream	0.193	0.300	0.931	0.093	0.500	0.156	0.277	0.875	0.587	0.548

Notes:

All regressions are estimated pooling data from the three follow-up surveys, using these weights, include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline and survey-wave fixed effects. Standard errors are robust and clustered at the respondent level. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Row-level p-values test (i) whether the effect of receiving a 36k grant differs from that of receiving a 9k grant in the multi-staged competition; (ii) whether the effect of receiving a 36k grant is four times that of receiving a 9k grant in the multi-staged competition; (iii) whether the effects of all three grant types are jointly equal to zero; and (iv) whether the effect of receiving 9k in the multi-staged competition differs from that in the streamlined process.

Table A4 continued

	Obtained a new Quality Accreditation (11)	Uses Internet for Email (12)	Uses Internet for Selling Products or Services to Clients (13)	Uses Internet for Advertising through Website (14)	Uses Internet for Advertising through Social Media (15)	Uses Internet to find out Information about Industry (16)	Uses Internet for Ordering Supplies (17)	Biz used Consulting Services in past 12 Months (18)	Taken Part in Training Program to Improve Biz Skills in past 12 Months (19)	Made Changes to Way Biz Operates as Result of Training (20)
36k-Multi	0.027* (0.014)	0.248*** (0.035)	0.197*** (0.036)	0.187*** (0.035)	0.208*** (0.035)	0.196*** (0.034)	0.209*** (0.036)	0.220*** (0.032)	0.197*** (0.036)	0.179*** (0.036)
9k-Multi	0.012 (0.014)	0.190*** (0.035)	0.110*** (0.037)	0.116*** (0.035)	0.135*** (0.035)	0.153*** (0.034)	0.146*** (0.037)	0.136*** (0.033)	0.175*** (0.036)	0.154*** (0.037)
9k-Stream	-0.003 (0.010)	0.219*** (0.033)	0.150*** (0.034)	0.083*** (0.029)	0.143*** (0.035)	0.193*** (0.032)	0.091*** (0.033)	0.139*** (0.028)	0.167*** (0.035)	0.154*** (0.034)
Observations Multi	1497	1497	1497	1497	1497	1497	1497	1497	1497	1497
Observations Stream	1103	1103	1103	1103	1103	1103	1103	1102	1102	1102
Control Multi Mean	0.032	0.494	0.513	0.238	0.567	0.608	0.353	0.236	0.409	0.382
Control Stream Mean	0.030	0.390	0.446	0.196	0.514	0.542	0.356	0.181	0.366	0.332
P-value: 36k = 9k-Multi	0.283	0.063	0.009	0.039	0.015	0.112	0.058	0.012	0.512	0.463
P-value: All Coefficients = 0	0.708	0.000	0.060	0.027	0.007	0.000	0.004	0.006	0.000	0.001
P-value: 36k = 4*9k-Multi	0.279	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
P-value: 9k-Multi = 9k-Stream	0.405	0.557	0.431	0.475	0.878	0.391	0.267	0.937	0.884	0.998

Notes:

All regressions are estimated pooling data from the three follow-up surveys, using these weights, include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline and survey-wave fixed effects. Standard errors are robust and clustered at the respondent level. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Row-level p-values test (i) whether the effect of receiving a 36k grant differs from that of receiving a 9k grant in the multi-staged competition; (ii) whether the effect of receiving a 36k grant is four times that of receiving a 9k grant in the multi-staged competition; (iii) whether the effects of all three grant types are jointly equal to zero; and (iv) whether the effect of receiving 9k in the multi-staged competition differs from that in the streamlined process.

Table A5: Heterogeneity by City

	Number of Employees (1)	More than 10 employees (2)	Profits (3)
36k	3.597*** (0.531)	0.143*** (0.029)	220** (102)
36k * City	0.735 (1.068)	0.021 (0.058)	65 (212)
	1.000	1.000	1.000
9k-Multi	2.294*** (0.531)	0.082*** (0.028)	181* (103)
9k-Multi * City	0.456 (0.892)	0.035 (0.052)	103 (233)
	1.000	1.000	1.000
9k-Stream	1.634*** (0.397)	0.041* (0.021)	-94 (78)
9k-Stream * City	0.082 (0.676)	-0.012 (0.037)	223* (131)
	1.000	1.000	1.000
City Dummy Multi	0.444 (0.569)	0.008 (0.034)	149 (142)
City Dummy Stream	0.012 (0.453)	0.020 (0.023)	-113 (94)
Number Obs Multi	1517	1517	1530
Number Obs Stream	1153	1153	1159
Control Mean From City Multi	3.743	0.114	535
Control Mean Not From City Multi	2.484	0.061	245
Control Mean From City Stream	1.989	0.049	211
Control Mean Not From City Stream	2.505	0.056	333
P-value: 36k*City = 9k-Multi*City	0.804	0.823	0.875
P-value: 9k-Multi*City = 9k-Stream*City	0.738	0.462	0.652

Notes:

This table reports heterogeneity in treatment effects by gender, interacting each treatment indicator with a dummy equal to one if the respondent lives in Nairobi, Mombasa or Kisumu. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Column (1) reports results for Number of Employees, which includes all workers employed by the business, excluding the owner, and is winsorized at the 99th percentile. Column (2) for Has 10 or More Employees, an indicator equal to one if the business employs at least 10 workers and Column (3) for Monthly Profits, winsorized at the 1st and 99th percentiles and deflated to real Kenyan shillings using 2024 prices, and then converted to US-dollars. All results are estimated using the pooled specification. Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values and survey-wave fixed effects. Standard errors are robust and clustered at the respondent level.

Sharpened q-values (Benjamini, Krieger, and Yekutieli (2006)) for the interaction terms are presented in square brackets and are computed jointly across all 36 heterogeneity tests (4 heterogeneity dimensions $\times$ 3 treatment arms $\times$ 3 outcomes). The heterogeneity dimensions are gender, business ownership at intake, city, and education.

Table A6: Heterogeneity by Education

	Number of Employees (1)	More than 10 employees (2)	Profits (3)
36k	3.731*** (0.792)	0.121*** (0.044)	309* (160)
36k * At least College Level	0.198 (0.983)	0.044 (0.054)	-88 (195)
9k-Multi	1.000 1.590* (0.882)	1.000 0.054 (0.047)	1.000 31 (145)
9k-Multi * At least College Level	1.139 (1.009)	0.053 (0.054)	243 (187)
9k-Stream	1.000 2.510*** (0.445)	1.000 0.104*** (0.027)	1.000 106 (78)
9k-Stream * At least College Level	-1.259** (0.612)	-0.100*** (0.035)	-183 (117)
At least College Level Dummy Multi	1.000 0.045 (0.557)	0.167 -0.002 (0.032)	1.000 61 (124)
At least College Level Dummy Stream	0.793** (0.358)	0.054*** (0.019)	132* (76)
Number Obs Multi	1517	1517	1530
Number Obs Stream	1153	1153	1159
Control Mean At Least College Multi	2.947	0.076	353
Control Mean Less Than College Multi	2.802	0.086	315
Control Mean At Least College Stream	2.743	0.071	360
Control Mean Less Than College Stream	1.455	0.017	146
p-val: 36k * At least College = 9k-Multi * At least College	0.436	0.882	0.120
p-val: 9k-Multi * At least College = 9k-Stream * At least College	0.042	0.017	0.054

Notes:

This table reports heterogeneity in treatment effects by gender, interacting each treatment indicator with a dummy equal to one if the respondent has at least college-level education. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. Column (1) reports results for Number of Employees, which includes all workers employed by the business, excluding the owner, and is winsorized at the 99th percentile. Column (2) for Has 10 or More Employees, an indicator equal to one if the business employs at least 10 workers and Column (3) for Monthly Profits, winsorized at the 1st and 99th percentiles and deflated to real Kenyan shillings using 2024 prices, and then converted to US-dollars. All results are estimated using the pooled specification. Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values and survey-wave fixed effects. Standard errors are robust and clustered at the respondent level.

Sharpened q-values (Benjamini, Krieger, and Yekutieli (2006)) for the interaction terms are presented in square brackets and are computed jointly across all 36 heterogeneity tests (4 heterogeneity dimensions × 3 treatment arms × 3 outcomes). The heterogeneity dimensions are gender, business ownership at intake, city, and education.

B. Business Plan Training

In the multi-stage selection process, a further process question of policy interest was whether it would help to provide training on putting together a business plan. KPMG provided an online set of seven modules that were adapted from existing sources such as the International Labour Organization's Start and Improve Your Business program and the Business Model Canvas. Among the 4,329 applicants selected to be eligible to submit business plans, 1,500 were randomly assigned to receive encouragement to participate in this online training. Among this group, 754 (50%) took up training, and 38% completed all modules. Since the training was online, the 2,749 who did not receive this encouragement could also participate, and 310 (11%) took it up, with 8% completing all modules.

In Table B1 we see that being randomly encouraged to participate in this training significantly increased the likelihood of submitting a business plan by about 3 percentage points.

However, it did not improve the performance in the competition itself, with no statistically significant effect on judges' scores conditional on submitting. This suggests that while the training was effective in encouraging participation, it did not translate into more successful submissions.

Table B1: Impact of Business Plan Training

	Submitted Business Plan		Top 750		Top 750 (Conditional on Plan Submission)		Judges Score	
	(1)	(2)	(3)	(4)	(5)	(6)	(3)	(4)
Invited to BPC Training	0.031** (0.014)	0.027* (0.014)	0.0099 (0.010)	0.0086 (0.010)	0.0068 (0.014)	0.0078 (0.013)	0.0980 (0.497)	0.1608 (0.490)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
p-value	0.028	0.051	0.330	0.397	0.613	0.559	0.844	0.743
N	4329	4329	4329	4329	3182	3182	3182	3182
R-sq.	0.001	0.022	0.000	0.015	0.000	0.013	0.000	0.031
Mean	0.735	0.735	0.116	0.116	0.157	0.157	56.427	56.427

Notes:

For each outcome we report results with and without controls. Controls included are strata and highest education level dummies. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively.

C. Robustness to Attrition

Table C1 provides response rates by treatment status. In both the multi-stage and streamlined processes, response rates were lower for the control group than for treated respondents. This was more pronounced in the multi-staged process, with control group response rates 15-22 percentage points lower than treatment, while in the streamlined process they were 3-8 percentage points lower, depending on the round. We discuss the reasons why this attrition occurred, the measures taken to reduce attrition, and the different robustness checks we carry out to argue that attrition is unlikely to be driving our main results. The main conclusion is that differential attrition is low between treatments, so should not affect our grant size experiment results. Moreover, those who attrit appear to be less likely to be operating firms, making our treatment versus control comparisons likely to be lower bounds on treatment effects.

Table C1: Survey Response Rates by Follow-up

	Any follow-up		1 Year Follow-up		2 Year Follow-up		3 Year Follow-up	
	Participation	Known Status	Participation	Known Status	Participation	Known Status	Participation	Known Status
36k Response Rate	95%	98%	89%	91%	75%	76%	68%	77%
9k-Multi Response Rate	91%	95%	86%	88%	70%	70%	68%	76%
9k-Stream Response Rate	94%	96%	88%	89%	76%	77%	72%	80%
Control Multi Response Rate	76%	91%	68%	72%	48%	52%	46%	76%
Control Stream Response Rate	91%	95%	85%	87%	68%	68%	66%	83%
Total Number of Observations	3351	3564	1040	1069	844	858	801	982
Multi Number of Observations	1962	2130	607	629	484	496	456	574
Stream Number of Observations	1389	1434	433	440	360	362	345	408
Overall Response Rate (All Groups)	89%	95%	83%	86%	68%	69%	64%	79%
P-value: 36k = Control Multi	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.832
P-value: 9k-Multi = Control Multi	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.916
P-value: 9k-Stream = Control Stream	0.008	0.450	0.237	0.409	0.046	0.028	0.147	0.356
P-value: 9k Multi = 9k Stream	0.017	0.450	0.422	0.777	0.157	0.105	0.329	0.280
P-value: 36k = 9k Multi	0.006	0.003	0.278	0.379	0.228	0.190	1.000	0.751

Known status is defined as those respondents that either completed the survey or those for whom we know if they had an operating business at the time of the survey.

Attrition was primarily a result of refusals, rather than inability to locate respondents. For example, in the third follow-up round, 32% of non-respondents explicitly refused to answer, 22% repeatedly postponed interviews (soft declines), and a further 24% blocked the survey firm's numbers after initial attempts. Only 22% of those not interviewed were due to respondents not being found or reached after tracking efforts. One reason for higher attrition in the third round (despite the measures described below) is that it took place in a period of nationwide protests in Kenya, initially triggered by opposition to proposed tax increases. There was criticism of the government and the International Monetary Fund for fiscal consolidation measures, which extended to the World Bank.¹⁶ This may have reduced overall willingness to participate in the survey. Furthermore, McKenzie (2019) showed that differential attrition is more common in experiments where the treatment group gets large grants and the control group does not, as is the case here.

¹⁶ Lawal, S. (2024, July 7). *What do the IMF and foreign debt have to do with Kenya's current crisis?* Al Jazeera. URL: <https://www.aljazeera.com/news/2024/7/7/why-are-kenyans-angry-with-the-imf> [last accessed: 06 February 2026]

Table C2: Balance Table among Respondents to the 3 Year Follow-up Survey at Baseline

	Control Multi mean	36K mean	9K-Multi mean	P-value: Control- Multi = 36k = 9k- Multi	9K- Stream mean	Control- Stream mean	P-value: 9k-Stream = Control- Stream	P-value: 9k-Multi = 9k-Stream
Strata Variables								
Female - Existing Business	0.12	0.14	0.15	0.744	0.23	0.25	0.648	0.058
Female - New Business	0.04	0.09	0.08	0.272	0.23	0.19	0.375	0.000
Male - Existing Business	0.50	0.44	0.49	0.483	0.27	0.30	0.456	0.000
Male - New Business	0.34	0.33	0.28	0.539	0.27	0.25	0.701	0.743
Owner characteristics								
Age	28.0	27.2	27.6	0.333	27.08	26.84	0.567	0.257
Secondary Education	0.15	0.18	0.12	0.297	0.26	0.24	0.691	0.001
Certificate-Level Education	0.06	0.07	0.07	0.982	0.05	0.05	0.850	0.501
University and College Education	0.73	0.67	0.75	0.271	0.67	0.67	0.905	0.091
Postgraduate education	0.06	0.09	0.07	0.720	0.02	0.02	0.901	0.042
Lived abroad at BL	0.054	0.007	0.043	0.104	0.007	0.030	0.139	0.044
Sector = Agriculture, farming and fishery	0.24	0.23	0.23	0.970	0.13	0.20	0.149	0.037
Sector = Wholesale and retail trade	0.20	0.20	0.19	0.975	0.26	0.23	0.676	0.229
Sector = Information and Communication	0.09	0.14	0.09	0.386	0.09	0.08	0.656	0.870
Sector = Manufacturing	0.03	0.13	0.14	0.038	0.12	0.10	0.696	0.706
Sector = Professional, scientific and technical activities	0.05	0.07	0.10	0.362	0.10	0.06	0.241	0.864
Employment Income	166	195	190	0.788	133	114	0.450	0.097
Primary outcomes at baseline								
Has Business	0.76	0.74	0.76	0.942	0.64	0.70	0.302	0.024
Conditional on having a Business:								
Has more than 1 Business	0.16	0.30	0.23	0.064	0.12	0.10	0.696	0.022
Sales	2364	2796	3497	0.508	1571	1473	0.832	0.023
Profit	642	458	627	0.617	476	390	0.610	0.411
Profit from All Businesses	686	582	638	0.912	481	393	0.602	0.401
Number of Employees	5.9	6.4	6.8	0.776	5.24	4.68	0.530	0.169
Percent of Paid Employees	0.78	0.90	0.86	0.131	0.86	0.80	0.280	0.933
Number of Hours Worked in Biz	42.0	40.9	41.6	0.962	34.83	35.28	0.903	0.081
Value of Inventory	3252	2895	2482	0.816	2046	2607	0.612	0.639
Value of Equipment	35825	39813	48871	0.296	21331	20350	0.812	0.000
Salay Paid to Self (from all Businesses)	170	136	243	0.005	137	146	0.766	0.003
Business Practices Index	0.74	0.75	0.75	0.904	0.74	0.70	0.109	0.694
Innovation and Training Index	0.49	0.52	0.52	0.440	0.44	0.45	0.771	0.003
Borrowed in the Past from a Microfinance Institution or Bank	0.27	0.27	0.23	0.776	0.22	0.17	0.260	0.858
Business Name Registered in Company Registrar	0.60	0.65	0.71	0.272				
Omnibus Test of Joint Orthogonality (P-value)				0.055			0.875	

Notes:

For age, information from intake is used to account for the fact that the two baselines did not take place at the same point of time. All monetary variables have been deflated to real Kenyan shillings using 2024 prices, and have then converted to US-dollars. Sales and Profit variables are reported for the last month. Number of Employees, Sales, Value of Inventory, have been winsorized at the 99 percentile. Value of equipment has been winsorized at the 95th percentile. Profits, Salary, and hours worked have been winsorized at the 1st and 99th percentiles.

These attrition rates occurred even after considerable field efforts to reduce attrition. Our protocols included: (i) reaching out to respondents a minimum of ten times over the phone at different dates and times to set interviews; (ii) conducting a minimum of three visits to individual respondents whenever the interview was not possible to be set over the phone using GPS coordinates and information on landmarks collected at baseline or in previous survey rounds; (iii) reducing the length of the survey questionnaire across rounds; (iv) implementing short surveys to be completed over the phone whenever respondents had no time to meet; (v) searching for respondents and their businesses on the web and on social

media (Facebook and LinkedIn) and physical tracking. Furthermore, in the third follow-up survey, an incentive scheme with nine prizes of KSh 64,000 (approximately \$500 each) was used to attempt to further increase responses.

Whenever interviews were not possible, survey teams sought to record whether the business was operating. This information was obtained either directly from declining participants or indirectly from knowledgeable contacts, including family members or individuals encountered during physical tracking who were familiar with the respondent's business status. Table C1 shows that this approach was very successful in closing the response gaps when it comes to defining business operating status. In the third follow-up round, business operating status is known for 76%-83% of firms, and the differences between treatment and control rates are only 1-3 percentage points and not statistically significant.

The sample of respondents to the third follow-up survey remains largely balanced on baseline characteristics (Table C2), although an omnibus test of joint orthogonality across all three treatment groups in the multi-stage experiment has $p=0.067$. We showed in Table 4 robustness to including these baseline controls, as well as to several other approaches for dealing with attrition. We use our data on known status for those not answering, as well as data across rounds to understand the selection into attrition. In the third follow-up round, 70% of those in the multi-stage process control group who completed the survey were operating a business, compared to only 51% of those who did not answer but whose status was known. This suggests that attrition should be leading to less successful firms being dropped on average, so that we are more likely to be at the top than the bottom of Lee bounds.

D. Measurement of Key Outcomes

Outcomes in nominal Kenyan shillings were deflated to real Kenyan shillings (July 2024 prices, i.e. the time of the third follow-up survey) using the CPI data from the Kenya National Bureau of Statistics (KNBS) and subsequently converted to USD also using the July 2024 exchange rate.

Key outcomes are measured as follows:

Business ownership: A binary variable measured by asking entrepreneurs whether they have a business. When an entrepreneur declined to participate in the full survey, we nevertheless attempted to collect this information where possible; if the entrepreneur could not be located, we sought this information from relatives or neighbors.

Total employment: The number of all workers in the firm, excluding the owner. The variable includes all types of workers, e.g. partners, full-time workers, family members who work for the company, unpaid workers, temporary workers, apprentices, managers, and co-owners

who work for the company. Coded as zero if the business does not exist and winsorized at the 99th percentile.

Firm of +10 workers: A binary variable taking the value one if total employment is 10 or more and zero otherwise.

Winsorized sales: Total sales reported for the previous month, and winsorized at the 99th percentile. Coded as zero if the business does not exist.

Winsorized profits: Total profits are reported for the previous month and winsorized at the 1st and 99th percentile to reduce the influence of outliers. Coded as zero if the business does not exist.

Business performance index: The average of the z-scores of winsorized sales and winsorized profits. For firms that are not operating, sales and profits are first coded as zero. Each outcome is then standardized using the control-group's mean and standard deviation. Coded as zero if the business does not exist.

Land and buildings: Measures the value of land and buildings, winsorized at the 99th percentile. Note that in the first follow-up this asked for value owned or rented, whereas the third follow-up only asked for the value of capital owned. So the two measures are not directly comparable to one another over waves.

Value of inventory: Current value reported of inventories, winsorized at the 99th percentile.

Capital stock (excluding land and buildings): Measures the firm's total capital stock as the sum of the monetary values of tools and utensils, machinery, animals, furniture, transport equipment, and other physical assets, winsorized at the 99th percentile.

Took out a loan in the past 12 months: Dummy taking the value one if the entrepreneur took out a loan for the business in the past 12 months in either of the three follow-up surveys.

Amount of loans taken out: Sum of loans the entrepreneur took out for the business across the three follow-up surveys.

Business practice index: the proportion of the following business practices employed, based on McKenzie and Woodruff (2017):

- Marketing Practices: Coded as one for each of the following that the business has done in the last 3 months
 - M1: Visited at least one of its competitor's businesses or websites to see what prices its competitors are charging. Coded as zero if entrepreneur says they have no competitors,

- M2: Visited at least one of its competitor's businesses or websites to see what products its competitors have available for sale. Coded as zero if entrepreneur says they have no competitors,
- M3: Asked existing customers whether there are any other products the customers would like the business to sell or produce,
- M4: Talked with at least one former customer to find out why former customers have stopped buying from this business. Coded as zero if entrepreneur does not have former customers,
- M5: Asked a supplier about which products are selling well in this business' industry
- M6: Used a special offer to attract customers,
- M7: Done any form of advertising,
- M8: Measured whether the advertisement was effective,
- M9: Made suggestions of other items a customer may buy.
- Buying and Stock Control Practices:
 - B1: Attempted to negotiate with a supplier for a lower price on raw material or goods purchased. Coded as 1 if the entrepreneur attempted to negotiate or does not have suppliers,
 - B2: Compared the prices or quality offered by an alternative supplier of raw material. Coded as one if the entrepreneur compared suppliers or reported having no suppliers,
 - B4: Keep stock of finished products. Coded as one if the entrepreneur keeps inventory and zero otherwise,
 - B5: An indicator ranging from 0 to 1 based on the time required to obtain goods after a stockout. Entrepreneurs able to replenish within one day receive a value of 1; those requiring more than one day but less than one week receive 0.66; those requiring one week receive 0.33; and those requiring one month or more, or that do not keep inventory, receive 0. Values are linearly scaled to lie between 0 and 1.
- Costing and Record-Keeping Practices: coded as one for each of the following
 - R1: Keeps written business records,
 - R2: Records every purchase and sale made by the business,
 - R3: Able to use records to see how much cash the business has on hand at any point in time,
 - R4: Uses records regularly to know whether sales of a particular product are increasing or decreasing from one month to another,
 - R5: Works out the cost to the business of each main product it sells,
 - R6: Knows which product/service makes the most profit per item selling,

- R7: Has a written budget, which states how much is owed each month for rent, electricity, equipment maintenance, transport, advertising, and other indirect costs to business,
- R8: Has records documenting that there exists enough money each month after paying business expenses to repay a loan in the hypothetical situation that this business wants a bank loan.
- Financial Planning Practices: Coded as one for each of the following
 - F1: Review the financial performance of their business and analyze where there are areas for improvement at least monthly,
 - F3: Compares their sales achieved to their target at least monthly,
 - F4: Has a budget of the likely costs their business will have to face over the next year.

Has a website: Dummy taking the value one if the business has a website and zero otherwise.

Innovation and training index: Average of the following different measures of innovative activities:

- Introduced a new product in past 12 months,
- Degree of product innovation (normalized indicator),
- Product or service copied by competitor in city. Coded as zero if entrepreneur says they have no competitors,
- Improved an existing product or service in the past 12 months,
- Introduced a new or improved business process in the past 12 months,
- Opened a new shop or location in the past 12 months,
- Trained staff in new areas for the business in the past 12 months,
- Licensed a new technology from another firm in the past 12 months,
- Brought in-house production activity that was previously outsourced,
- Introduced a new in-house training program in the past 12 months,
- Obtained a new quality accreditation,
- Uses internet for email,
- Uses internet for selling products or services to clients,
- Uses internet for advertising through website,
- Uses internet for advertising through social media,
- Uses internet to find out information about industry,
- Uses internet for ordering supplies,
- Used consulting service over the past 12 months,
- Taken part in a training program to improve business skills in the past 12 months,

- Made changes to way the business operates as result of the training.

Employment income: Income from work in the last month, including wage employment (paid by salary or commission), work on a household member's farm or agricultural land, helping in a business owned or run by a household member, and work as a volunteer, intern, or apprentice.

Total income: Sum of employment income plus profit from all businesses combined, winsorized at the 1st and 99th percentile.

E. Impacts on Secondary Firm Performance Outcomes

Table E1 shows impacts on firm performance conditional on operation. Table E2 shows impacts on total income from both wage employment and business profits, as well as impacts on log sales per worker conditional on operating.

Table E1: Intensive Margin - Impacts on Firm Performance

		1 Year Follow-up (1)	2 Year Follow-up (2)	3 Year Follow-up (3)	(1)=(2)=(3)	Pooled (4)
	36k	885** (434)	617 (486)	663.744 (585)	0.907	729** (364)
	9k-Multi	378 (485)	656 (547)	616 (674)	0.918	529 (419)
	9k-Stream	24 (365)	37 (421)	-423 (318)	0.555	-77 (255)
Monthly Sales	Observations Multi	568	439	357		1364
	Observations Stream	361	288	247		896
	Control Multi Mean	2107	1778	2074		2002
	Control Stream Mean	1783	1566	1515		1636
	P-value: 36k = 9k-Multi	0.260	0.936	0.938		0.619
	P-value: All Coefficients = 0	0.235	0.582	0.364		0.238
	P-value: 36k = 4*9k-Multi	0.720	0.304	0.461		0.368
	P-value: 9k-Multi = 9k-Stream	0.560	0.370	0.164		0.217
	36k	49 (143)	214 (142)	248 (156)	0.583	149 (107)
	9k-Multi	48 (142)	205 (160)	157 (161)	0.744	124 (114)
	9k-Stream	-100 (133)	15 (107)	-169* (98)	0.437	-77 (86)
Monthly Profits	Observations Multi	567	436	357		1360
	Observations Stream	360	287	247		894
	Control Multi mean	505	452	523		494
	Control Stream mean	488	341	442		430
	p-value: 36k = 9k-Multi	0.995	0.947	0.550		0.808
	P-value: All Coefficients = 0	0.872	0.481	0.141		0.407
	P-value: 36k = 4*9k-Multi	0.770	0.281	0.509		0.394
	p-value: 9k-Multi = 9k-Stream	0.447	0.322	0.085		0.159
	36k	0.115 (0.099)	0.188 (0.138)	0.227 (0.173)	0.817	0.165* (0.097)
	9k-Multi	0.061 (0.110)	0.189 (0.158)	0.170 (0.193)	0.758	0.130 (0.111)
	9k-Stream	-0.056 (0.099)	0.021 (0.129)	-0.188 (0.117)	0.458	-0.059 (0.082)
Business Performance Index	Observations Multi	568	439	357		1364
	Observations Stream	361	288	247		896
	Control Multi mean	0.081	0.123	0.276		0.145
	Control Stream mean	0.122	0.142	0.211		0.155
	p-value: 36k = 9k-Multi	0.586	0.990	0.744		0.732
	P-value: All Coefficients = 0	0.644	0.558	0.227		0.314
	P-value: 36k = 4*9k-Multi	0.743	0.308	0.511		0.381
	p-value: 9k-Multi = 9k-Stream	0.431	0.410	0.113		0.172

Notes:

Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. The pooled specification includes survey-wave fixed effects. Standard errors are robust and, in the pooled specification, clustered at the respondent level. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. The sample is limited to respondents who operate a business at the time of surveying. Column 4 reports p-values for tests of equality across the first, second, and third follow-up rounds. Row-level p-values test (i) whether the effect of receiving a 36k grant differs from that of receiving a 9k grant in the multi-stage competition; (ii) whether the effect of receiving a 36k grant is four times that of receiving a 9k grant in the multi-stage competition; (iii) whether the effects of all three grant types are jointly equal to zero; and (iv) whether the effect of receiving 9k in the multi-stage competition differs from that in the streamlined process.

Monthly Sales are sales reported from the last month winsorized at the 99th percentile and deflated to real Kenyan shillings using 2024 prices, and then converted to US-dollars. **Monthly Profits** are winsorized at the 1st and 99th percentiles; all other transformations are the same. **Business Performance Index** is measured as an index averaging standardized (z-scored) monthly sales and profits. Each component is standardized using the control group mean and standard deviation, and the index is computed as the mean of these standardized variables.

Table E2: Additional Impacts on Total Income and on Firm Performance

		1 Year Follow-up	2 Year Follow-up	3 Year Follow-up	(1)=(2)=(3)	Pooled
		(1)	(2)	(3)		(4)
	36k	43 (125)	288** (128)	347*** (103)	0.153	223** (89)
	9k-Multi	154 (132)	330** (143)	310*** (105)	0.577	260*** (98)
	9k-Stream	-89 (108)	19 (82)	-9 (70)	0.720	-28 (65)
Total Income	Observations Multi	612	496	574		1682
	Observations Stream	431	362	408		1201
	Control Multi mean	646	530	362		505
	Control Stream mean	542	429	365		447
	p-value: 36k = 9k-Multi	0.360	0.759	0.739		0.698
	P-value: All Coefficients = 0	0.538	0.074	0.003		0.026
	P-value: 36k = 4*9k-Multi	0.224	0.048	0.022		0.022
	p-value: 9k-Multi = 9k-Stream	0.155	0.060	0.012		0.014
	36k	-0.106 (0.138)	0.004 (0.172)	-0.092 (0.174)	0.873	-0.073 (0.113)
	9k-Multi	-0.084 (0.139)	0.142 (0.176)	-0.084 (0.176)	0.545	-0.018 (0.114)
	9k-Stream	-0.293* (0.152)	-0.422** (0.167)	0.040 (0.170)	0.125	-0.245** (0.116)
Sales per Worker	Observations Multi	455	361	273		1089
	Observations Stream	268	226	174		668
	Control Multi mean	5.517	5.500	5.862		5.604
	Control Stream mean	5.477	5.683	5.458		5.536
	p-value: 36k = 9k-Multi	0.862	0.294	0.957		0.569
	P-value: All Coefficients = 0	0.229	0.054	0.947		0.172
	P-value: 36k = 4*9k-Multi	0.638	0.344	0.690		0.997
	p-value: 9k-Multi = 9k-Stream	0.311	0.020	0.614		0.163

Notes:

Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. The pooled specification includes survey-wave fixed effects. Standard errors are robust and, in the pooled specification, clustered at the respondent level. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. The sample is limited to respondents who operate a business at the time of surveying. Column 4 reports p-values for tests of equality across the first, second, and third follow-up rounds. Row-level p-values test (i) whether the effect of receiving a 36k grant differs from that of receiving a 9k grant in the multi-stage competition; (ii) whether the effect of receiving a 36k grant is four times that of receiving a 9k grant in the multi-stage competition; (iii) whether the effects of all three grant types are jointly equal to zero; and (iv) whether the effect of receiving 9k in the multi-stage competition differs from that in the streamlined process.

Total Income is the sum of employment income plus profit from all businesses combined winsorized at the 1st and 99th percentiles; all other transformations are the same. **Sales per worker** is the log of last month's sales divided by the number of employees and is only defined for those running a business. The decline in observations across follow-up rounds reflects both business closure and firms operating without any employees.

F. Additional Tables and Figures on Mechanisms

Figure F1: Correlation Application Score vs Multi-stage judge Score

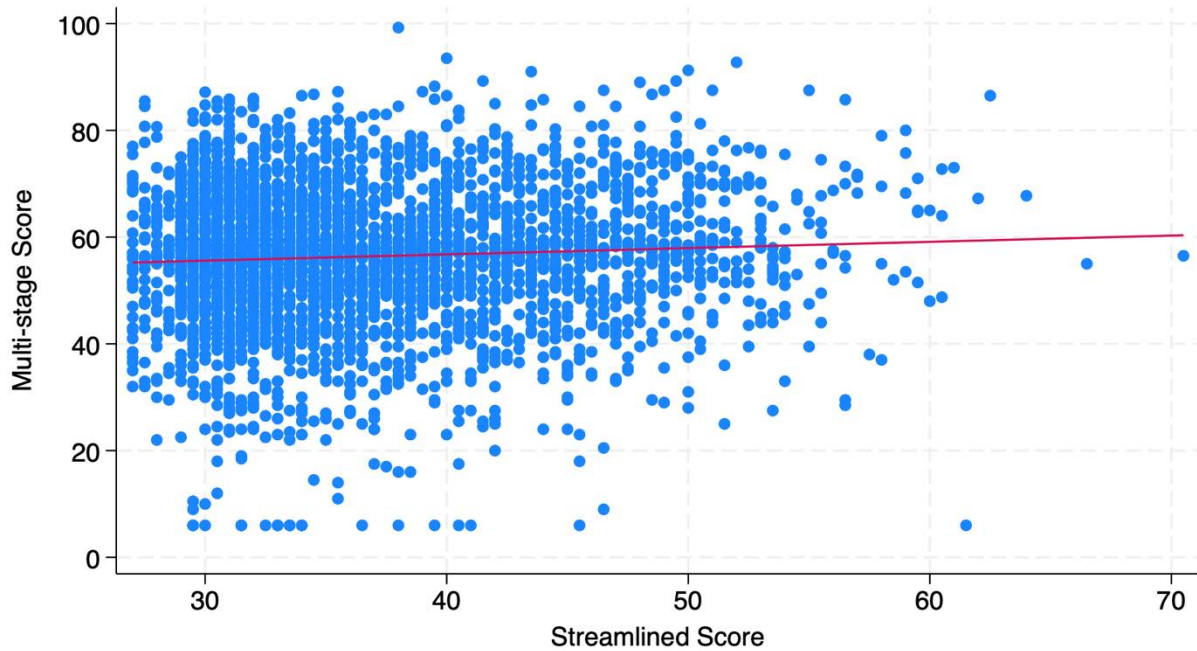


Table F1: Heterogeneity in Impacts of 9k-Stream for those with Above vs Below Median Application Scores

	3 Year Follow-up survey			Pooled Specification		
	Number of Employees (1)	More than 10 employees (2)	Profits (3)	Number of Employees (4)	More than 10 employees (5)	Profits (6)
9k-Stream	0.718 (0.517)	0.022 (0.029)	-48 (63)	1.831*** (0.444)	0.065*** (0.023)	17 (65)
9k-Stream * Above	-0.260 (0.726)	-0.025 (0.045)	-57 (119)	-0.338 (0.635)	-0.053 (0.035)	-67 (127)
Dummy Above	0.428 (0.557)	0.028 (0.030)	107 (111)	0.109 (0.446)	0.019 (0.021)	117 (99)
Stream Observations	372	372	371	1153	1153	1159
Stream Control Mean	2.022	0.049	274	2.337	0.054	292

Notes:

Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. The pooled specification includes survey-wave fixed effects. Standard errors are robust and, in the pooled specification, clustered at the respondent level. Above is a dummy variable taking the value 1 if the respondent scored above the median in their application. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. **Number of Employees** includes all workers employed by the business, excluding the owner, and is winsorized at the 99th percentile. **Has 10 or More Employees** is an indicator equal to one if the business employs at least 10 workers. **Monthly Profits** are winsorized at the 1st and 99th percentiles; all other transformations are the same.

Table F2: Heterogeneity in Impacts for the Multi-stage Top 750 by Judges' scores

	3 Year Follow-up survey			Pooled Specification		
	Number of Employees (1)	More than 10 employees (2)	Profits (3)	Number of Employees (4)	More than 10 employees (5)	Profits (6)
36k	2.333*** (0.651)	0.076** (0.036)	188** (96)	3.185*** (0.629)	0.126*** (0.033)	13 (87)
36k * Above	0.734 (1.137)	0.052 (0.065)	234 (221)	1.183 (0.921)	0.045 (0.051)	412** (176)
9k-Multi	1.620** (0.648)	0.053 (0.035)	128 (109)	2.025*** (0.568)	0.079*** (0.029)	83 (105)
9k-Multi * Above	0.902 (1.079)	0.071 (0.064)	198 (222)	0.435 (0.842)	0.016 (0.046)	194 (189)
Dummy Above	0.534 (0.593)	0.041 (0.037)	-44 (131)	0.184 (0.540)	0.034 (0.028)	-60 (117)
Observations Multi	508	508	507	1584	1584	1597
Control Multi Mean	1.993	0.054	291	2.947	0.081	352
P-value: 36k = 9k (Below)	0.365	0.574	0.626	0.059	0.163	0.462
P-value: 36k = 9k (Above)	0.592	0.947	0.646	0.014	0.084	0.378
P-value: 36k = 9k-Multi (joint)	0.579	0.852	0.800	0.010	0.089	0.523

Notes:

Regressions include randomization strata and the baseline value of the outcome, as well as an indicator for missing baseline values. The pooled specification includes survey-wave fixed effects. Standard errors are robust and, in the pooled specification, clustered at the respondent level. Above is a dummy variable taking the value 1 if the respondent scored above the median in their application. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively. **Number of Employees** includes all workers employed by the business, excluding the owner, and is winsorized at the 99th percentile. **Has 10 or More Employees** is an indicator equal to one if the business employs at least 10 workers. **Monthly Profits** are winsorized at the 1st and 99th percentiles; all other transformations are the same.

Table F3: Baseline Correlates of Judges' Score in the Multi-stage Sample

Variable	Baseline judge score
Strata: Female, new business	-0.809 (0.938)
Strata: Male, existing business	-0.483 (0.575)
Strata: Male, new business	-0.242 (0.678)
Age	-0.004 (0.054)
Education = Certificate	-0.692 (0.994)
Education = University/College	0.834 (0.632)
Education = Postgraduate	0.798 (0.951)
Sector = Agriculture	0.738 (0.689)
Sector = Retail	1.284 (0.833)
Sector = Information/Communication	0.333 (0.778)
Sector = Manufacturing	1.508 (0.960)
Sector = Technology	-0.782 (0.826)
Number of Businesses	-0.403 (0.462)
Has more than 1 Business	0.627 (0.790)
Sales	0.000*** (0.000)
Profit	-0.000 (0.000)
Profit from all Businesses	-0.001 (0.002)
Employees	-0.011 (0.027)
Percentage of Paid Workers	-0.137 (0.634)
Hours Worked on Business	0.004 (0.008)
Inventory Value	0.000 (0.000)
Equipment Value	-0.000 (0.000)
Business Salary	0.000 (0.001)
Employment Income	-0.001 (0.002)
Total Income	0.001 (0.002)
Business Practices Index	-2.970 (1.992)
Innovation and Training Index	2.813 (1.803)
Expects Life Improvement in 5 Years	1.078 (1.522)
Confidence	1.859 (1.579)
Borrowed in the Past from a Microfinance Institution or Bank	-0.122 (0.553)
Lived Abroad at Baseline	-2.167 (1.399)

Business Name Registered in Company Registrar	1.092**
	(0.546)
Streamlined Score	0.047*
	(0.026)
N	750
R-squared	0.093

Notes:

For age, information from intake is used to account for the fact that the two baselines did not take place at the same point of time. All monetary variables have been deflated to real Kenyan shillings using 2024 prices, and have then converted to US-dollars. Sales and Profit variables are reported for the last month. Number of Employees, Sales, Value of Inventory, have been winsorized at the 99 percentile. Value of equipment has been winsorized at the 95th percentile. Profits, Salary, and hours worked have been winsorized at the 1st and 99th percentiles.